



FACULTY OF INFORMATION TECHNOLOGY AND ELECTRICAL ENGINEERING

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**TOWARDS BEHAVIOUR-AWARE MULTIMODAL
VIDEO SUMMARIZATION: INTEGRATING
VISUAL, AUDIO, AND TEXTUAL CUES FOR
HUMAN-CENTRIC CONTENT ANALYSIS**

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ABSTRACT

Video summarization models for human-centric content face critical challenges in capturing behavioural cues that convey essential interpersonal communication signals, as existing approaches either rely solely on visual features or employ basic multimodal combinations that neglect synchronized behavioural signals across modalities. This thesis proposes a behaviour-aware multimodal video summarization pipeline that integrates visual, audio, and textual modalities to identify moments of communicative significance through synchronized behavioural signals including gestures, facial expressions, and vocal prosody. The pipeline employs two complementary approaches: a transformer-based framework with cross-modal attention mechanisms and a heuristic approach utilizing multimodal *bonus words*, terms emphasized across multiple modalities that improve semantic relevance and expressive clarity of summaries. To address dataset scarcity, a scalable pseudo-ground truth generation pipeline using large language models (LLMs) enables automated evaluation without human annotations. Experimental validation on the ChaLearn First Impressions dataset demonstrates substantial performance improvements over existing state-of-the-art methods across multiple evaluation metrics. Comprehensive analysis confirms that multimodal integration significantly outperforms individual modalities, while temporal alignment emerges as crucial for maintaining narrative coherence and summary quality. This work contributes a novel methodology for preserving both semantic content and behavioural expressiveness in video summarization, enabling applications in educational technology, content analysis, and human-computer interaction.

Keywords: multimodal feature integration, cross-modal attention, behavioural cues detection, pseudo-ground truth generation, temporal alignment, multimedia content analysis, human-computer interaction.

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TIIVISTELMÄ

Ihmiskeskeisen sisällön videoyhteenvetomallit kohtaavat kriittisiä haasteita sellaisten käyttäytymisvihjeiden tallentamisessa, jotka välittävät olennaisia ihmisten välisiä viestintäsignaaleja, sillä nykyiset lähestymistavat joko luottavat pelkästään visuaalisiin ominaisuuksiin tai käyttävät perusmultimodaalisia yhdistelmiä, jotka jättävät huomiotta eri modaaliteettien synkronoidut käyttäytymissignaalit. Tässä opinnäytetyössä ehdotetaan käyttäytymistietoista multimodaalista videoyhteenvetoprosessia, joka yhdistää visuaaliset, audio ja tekstimuodot tunnistamaan kommunikatiivisesti tärkeitä hetkiä synkronoitujen käyttäytymissignaalien avulla, mukaan lukien eleet, ilmeet ja ääniprosodiat. Prosessissa käytetään kahta toisiaan täydentävää lähestymistapaa: muuntajapohjainen kehys, jossa on monimuotoisia huomiomekanismeja, ja heuristinen lähestymistapa, jossa käytetään multimodaalisia *bonussanatermejä*, jotka korostuvat useissa modaaliteeteissa ja jotka parantavat tiivistelmien semanttista merkitystä ja ilmaisullista selkeyttä. Tietojoukon niukkuuden korjaamiseksi skaalautuva pseudopohjatotuusprosessi, joka käyttää suuria kielimalleja (LLMs), mahdollistaa automaattisen arvioinnin ilman ihmisen annotointia. ChaLearn First Impressions-tietojoukon kokeellinen validointi osoittaa merkittäviä suorituskyvyn parannuksia nykyisiin uusimpiin menetelmiin verrattuna useissa arviointimitoissa. Kattava analyysi vahvistaa, että multimodaalinen integraatio on huomattavasti parempi kuin yksittäiset modaaliteetit, kun taas ajallinen kohdistaminen on ratkaiseva tekijä kerronnan johdonmukaisuuden ja yhteenvedon laadun ylläpitämisessä. Tämä työ edistää uudenlaista metodologiaa sekä semanttisen sisällön että käyttäytymisen ilmaisukyvyn säilyttämiseksi videoiden yhteenvedoissa, mikä mahdollistaa sovellukset koulutusteknologiassa, sisältöanalyysissä sekä ihmisen ja tietokoneen vuorovaikutuksessa.

Avainsanat: multimodaalisten ominaisuuksien integrointi, monimuotoinen huomio, käyttäytymisvihjeiden havaitseminen, pseudopohjatotuuden generointi, ajallinen kohdistus, multimediasisältöanalyysi, ihmisen ja tietokoneen vuorovaikutus.

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FOREWORD

The pursuit of this research has been an intellectually stimulating journey that challenged my understanding of multimodal systems and reshaped my perspective on human-centered technology. What began as curiosity about how machines might better understand human communication evolved into a comprehensive exploration of behaviour-aware video summarization.

I owe tremendous gratitude to my supervisors, Dr. Mourad Oussalah and Dr. Janne Heikkilä, at the Center for Machine Vision and Signal Analysis (CMVS) at the University of Oulu. Your unwavering support, guidance and expertise provided the perfect foundation for this interdisciplinary work, while the freedom to explore unconventional approaches, combined with rigorous scientific standards, fundamentally shaped my research methodology.

I am profoundly grateful to my family, whose love, support, and unwavering belief in my abilities have been my constant source of strength. Your encouragement during the most challenging phases of this research, your understanding during long hours of work, and your celebration of every small milestone have made this journey meaningful. You have been my anchor throughout this academic voyage.

The vibrant research community at CMVS deserves special recognition. Our discussions about transformer architectures, debates over evaluation metrics, and collaborative troubleshooting sessions transformed what could have been an isolating process into an engaging intellectual adventure.

I gratefully acknowledge the financial support from the University of Oulu and the Finnish Research Council Flagship 7 Hybrid Intelligence, funded through the Finnish Research Council Digital Water Flagship project. This support provided essential computational resources and the freedom to pursue innovative research directions.

This thesis embodies the power of curiosity-driven research, interdisciplinary collaboration, and perseverance in facing complex challenges. I hope this work opens new avenues for understanding how technology can better capture and preserve the richness of human communication.

Oulu, June 1st, 2025

Md Moinul Islam

LIST OF ABBREVIATIONS AND SYMBOLS

AI	Artificial Intelligence
ASR	Automatic Speech Recognition
BLEU	Bilingual Evaluation Understudy
BS	BERTScore
CLIP	Contrastive Language-Image Pre-training
CM-Attn	Cross-modal attention
CNNs	Convolutional Neural Networks
CS	CLIPScore
dB	Decibels
eGeMAPS	extended Geneva Minimalistic Acoustic Parameter Set
EOS	End-of-Sequence
FFN	Feed-Forward Network
FCSN	Fully Convolutional Sequence Networks
fps	Frames per second
GANs	Generative Adversarial Networks
GPE	Geo-Political Entity
HuBERT	Hidden-unit BERT
IDF	Inverse Document Frequency
IoU	Intersection-over-Union
LLMs	Large Language Models
LR	Length Ratio
LSTMs	Long Short-Term Memory networks
MA-LMM	Memory-Augmented Large Multimodal Model
MFA	Montreal Forced Aligner
MFST	Multimodal Frame-Scoring Transformer
ms	Millisecond
NaN	Not a number
NCC	Normalized cross-correlation
NER	Named Entity Recognition
NLSS	Near-Lossless Semantic Summarization
NRC	National Research Council
ORG	Organization
pGT	Pseudo-ground truth
RL	Reinforcement Learning
RNNs	Recurrent Neural Networks
RoBERTa	Robustly Optimized BERT Pretraining Approach
ROUGE	Recall-Oriented Understudy for Gisting Evaluation
RQs	Research Questions
SDG	Sustainable Development Goals
SOS	Start-of-Sequence
SOTA	State-of-the-art
SVD	Singular Value Decomposition
SVMs	Support Vector Machines
TF-IDF	Term Frequency-Inverse Document Frequency

ViT	Vision Transformer
VLMs	Vision-Language Models
YAAPT	Yet Another Algorithm for Pitch Tracking
τ_{window}	Alignment window
$\hat{A}$	Audio encoded features
w_a	Band-specific weights
ϵ	Buffer time in seconds
D	Corpus
d	Document
d_{model}	Dimension
δ	Diversity factor
$E(f)$	Energy density at frequency (f)
E_b	Energy in critical bands
F_0	Fundamental frequency/Pitch
I_i	Image embeddings
τ	Kendall's correlation coefficient
μ	Mean value
N	Number of frames
P_w	Positional weighting
Θ	Pre-trained parameter
θ	Regularization parameter
Δ	Sampling interval
ϕ	Scaling factor
ρ	Spearman's correlation coefficient
σ	Standard deviation
ω	Temperature parameter
t_r	Temporal decay rate
T_i	Text embeddings
$\hat{T}$	Textual encoded features
λ	Threshold parameter
M	Tracked landmarks
$L \ x$	Transformer layers
$\hat{V}$	Visual encoded features
x_t	Waveform segment
t_w	Word timestamp

1. INTRODUCTION

The exponential growth of video content on platforms such as YouTube, LinkedIn, news and educational repositories has revolutionized information dissemination, with billions of hours of footage uploaded annually [1]. This deluge necessitates automated video summarization tools that create concise representations preserving critical content, enabling efficient navigation and accessibility for applications in education, journalism, and professional communication [2]. As shown in Figure 1, the generic framework of summarizing videos usually involves analyzing the original video, identifying important content, and generating a condensed representation.

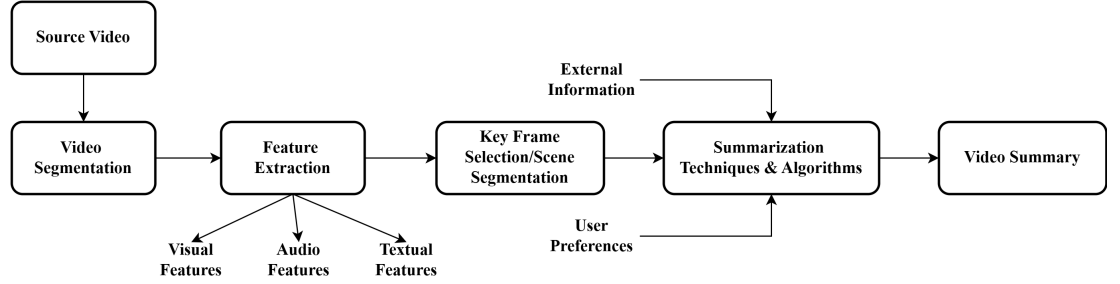


Figure 1. **The generic framework of the video summarization process [3].** This diagram illustrates a structured approach to video summarization, showing how videos are condensed into concise summaries while preserving important information.

Traditional summarization methods, which rely on visual features such as keyframes or scene transitions, often fail to capture the nuanced interplay of visual, auditory, and textual modalities that define human-centric videos, such as interviews or lectures [4]. This thesis proposes a behaviour-aware multimodal video summarization framework that integrates these modalities, prioritizing communicative signals such as gestures, facial expressions, and vocal prosody to produce contextually rich summaries. By integrating both heuristic and transformer-based architectures and large language models (LLMs), this thesis addresses challenges such as modality misalignment, dataset scarcity, and evaluation limitations, drawing inspiration from recent advances in video-text alignment and LLM-driven summarization [5, 6, 7]. This chapter outlines the motivation, research questions, objectives, and contributions, setting the stage for a comprehensive exploration of behaviour-aware summarization.

1.1. Motivation

The development of a behaviour-aware multimodal video summarization framework is driven by urgent practical needs and technological opportunities. In fields such as journalism, professionals require rapid summaries of lengthy news footage to identify critical events, while educators seek condensed lecture videos to enhance accessibility for diverse learners, including those with time constraints or disabilities [2]. Similarly, corporate decision-makers rely on concise video summaries to extract actionable insights from webinars or training sessions, where non-verbal cues such as a speaker’s emphatic gesture or tone shift often highlight key points [8]. Current summarization

systems, however, struggle to deliver coherent, contextually rich outputs, particularly for long-form or diverse video content, due to limited datasets and evaluation metrics that prioritize frame selection over narrative flow [9]. This gap underscores the need for a framework that captures communicative signals to produce summaries that resonate with human interpretation.

Technological advancements further motivate this research. Recent breakthroughs in LLMs and cross-modal attention mechanisms [10, 11] enable sophisticated processing of multimodal data, offering the potential to model complex interactions between visual, auditory, and textual cues. For instance, integrating gesture detection with prosodic analysis can highlight moments of emphasis in interviews, yet existing frameworks rarely exploit these capabilities fully [7, 8]. By addressing challenges such as temporal misalignment and the lack of annotated datasets [12, 6], this thesis aims to leverage these tools to create summaries that are not only concise but also emotionally and contextually accurate. Such a system promises to enhance user satisfaction, support real-time applications such as automated content curation, and promote inclusive access to multimedia content, aligning with goals for equitable information access.

1.2. Research Questions and Goals

This thesis aims to design and implement a robust framework that integrates visual, auditory, and textual modalities to produce concise, contextually rich video summaries, addressing both technological and practical needs in fields such as education, journalism, and content creation. To achieve this, the thesis investigates the feasibility of leveraging state-of-the-art multi-modal processing techniques, including LLMs and cross-modal attention mechanisms, to overcome limitations in existing summarization approaches, such as dataset scarcity and inadequate evaluation metrics. The primary research question guiding this thesis is how to design multimodal integration frameworks that effectively capture synchronized behavioural signals across visual, audio, and textual modalities for generating contextually-aware video summaries. To address this question, the thesis is structured around specific research questions (RQs) that break down the problem into manageable components, each contributing to the development and evaluation of the proposed framework:

RQ1: How can behavioural cues, such as gestures, facial expressions, and vocal prosody, be extracted and fused with semantic content to identify salient moments in human-centric videos?

This question explores feature extraction techniques, such as MediaPipe [13] for facial movement tracking, DeepFace [14] for emotion transitions, and CLIP [15] for visual embeddings, YAAPT [16], OpenSMILE [17] and HuBERT [18] for prosodic feature analysis and audio embeddings, RoBERTa [19] for textual embeddings and fusion methods such as cross-modal attention [11] to prioritize moments of communicative significance, addressing challenges such as modality misalignment [5, 6, 8].

RQ2: How does aligning transcript data with timestamped prosodic and visual cues improve the coherence and precision of both video and text-based summaries?

Temporal alignment of transcripts with audio and visual signals can enhance the narrative coherence of summaries, particularly for narrated videos [20]. This question investigates methods for timestamped multi-modal alignment, such as cross-modal attention mechanisms, to ensure that generated summaries accurately reflect the video’s temporal dynamics and semantic content.

RQ3: How can LLMs be adapted to generate pseudo-ground truth summaries that incorporate behavioural and multimodal cues when human annotation is unavailable?

This question investigates techniques for adapting LLMs to create scalable reference summaries through temporal prompt tuning and behavioural cue integration, building on LLM-driven pseudo-ground truth generation approaches [6, 10]. The focus lies in utilizing LLMs to produce automated reference summaries that preserve both semantic importance and behavioural salience, addressing the critical challenge of dataset scarcity in behaviour-aware video summarization evaluation.

RQ4: What evaluation metrics best assess the quality of behaviour-aware multimodal summaries when evaluated against LLM-generated summaries?

Traditional evaluation frameworks prove inadequate for capturing multimodal summary coherence and behavioural relevance due to inherent segmentation biases and limited scope [9]. This question addresses the development of comprehensive evaluation methodologies that integrate complementary text-based metrics (ROUGE, BERTScore) with video-specific assessments (CLIPScore) and temporal consistency measures (Kendall’s τ , Spearman’s ρ), establishing a robust framework for assessing behaviour-aware summarization quality against pseudo-ground truth references across multiple evaluation dimensions.

Goals. These research questions collectively aim to advance this research by addressing critical challenges in multi-modal integration, model adaptability, and evaluation robustness. The goals of this thesis are to develop a scalable and automated multimodal summarization framework that seamlessly integrates state-of-the-art techniques and behaviour-aware techniques, and evaluate its performance on LLM generated summaries and against established benchmarks. By answering these questions, the thesis aims to contribute to theoretical and the practical advancements in video summarization.

1.3. Objectives and Contributions

We present the objectives and contributions of this thesis, which develops a multi-modal video summarization framework that integrates visual, auditory, and textual modalities to create behaviour-aware summaries. By capturing human behavioural cues, such as head movements, emotions, loudness, pitch, and tonality, the framework aims to produce concise and contextually rich summaries.

Objectives. The objectives address the research questions (RQs) in Section 1.2. By tackling challenges such as modality integration and evaluation limitations, the thesis aims to improve summarization techniques for real-world applications.

1. To design a framework that integrates visual, auditory, and textual modalities, emphasizing behavioural cues such as gestures, facial expressions, and vocal prosody, to produce summaries that capture both semantic and emotional significance.
2. To implement a transformer-based architecture with cross-modal attention to model temporal and communicative interactions, ensuring alignment and coherence across modalities, inspired by video-text alignment techniques [5, 21, 22].
3. To develop a pseudo-ground truth generation pipeline using LLMs to address dataset scarcity, enabling robust training and evaluation of behaviour-aware summaries [6, 23].
4. To evaluate the framework on a dataset of human-centric videos, such as interviews, using a combination of text and video-based metrics to assess the summary quality.
5. To analyze the societal impacts, including accessibility benefits and potential biases, and to guide future research in ethical and inclusive summarization.

Contributions. This thesis offers the following contributions to the field of video summarization:

1. A novel behaviour-aware multimodal framework that prioritizes communicative signals, enhancing summary coherence and emotional fidelity in human-centric videos, building on cross-modal fusion techniques.
2. A cross-modal attention mechanism that dynamically aligns visual, auditory, and textual cues, addressing temporal misalignment and improving narrative flow.
3. An LLM-driven pseudo-ground truth generation pipeline, adapting pretraining strategies to create scalable and automated video summaries as references, overcoming the lack of annotated summaries [6, 23].
4. Comprehensive evaluation results on benchmark datasets, demonstrating superior performance compared to existing state-of-the-art methods, using metrics such as, ROUGE [24], BERTScore [25], F1-Score and CLIPScore [26].

1.4. Thesis Structure

The content of the thesis is organized as follows:

Section 1 introduces the need for behaviour-aware multimodal video summarization, outlining the context, motivation, research questions, objectives, and contributions.

Section 2 reviews state-of-the-art methods, focusing on unimodal, multimodal, and LLM-driven approaches, identifying gaps addressed by this thesis.

Section 3 details the proposed methodology, including dataset selection, behavioural cues extraction, heuristic and transformer architectures, temporal alignment, summary generation, and pseudo-ground truth generation using LLMs.

Section 4 presents the experimental setup, evaluation metrics, and results, comparing the proposed frameworks against baselines and analyzing their effectiveness.

Section 5 discusses key findings of both proposed approaches, limitations, and proposes future directions for enhancing behaviour-aware summarization across diverse domains.

Section 6 summarizes the content of the thesis, and establishes the contribution of behaviour-aware multimodal video summarization to the field.

1.5. Author's Contributions and the Role of Artificial Intelligence

The author independently formulated the research questions, designed and implemented the behaviour-aware multimodal framework, and evaluated with both text and video-based metrics against baseline methods, and conducted a comprehensive literature review to contextualize the work. The implementation, including dataset preprocessing, model architecture, and evaluation, was executed by the author under the guidance of the thesis supervisor. The writing, including the articulation of arguments and interpretation of findings, reflects the author's original contributions.

Artificial intelligence (AI) tools were employed to enhance the presentation and documentation of this thesis while preserving the integrity of original research contributions. Specifically, Grok 3 (developed by xAI) assisted in refining the clarity and readability of written content across multiple drafts, while Claude (developed by Anthropic) provided support for debugging code snippets, resolving LaTeX formatting issues, and structuring tabular data. All AI-generated suggestions underwent rigorous evaluation and extensive revision by the author to ensure alignment with academic standards and research objectives. The fundamental research contributions, including the conceptual framework development, experimental design, data analysis, and scholarly conclusions, represent entirely original work by the author, with AI tools serving exclusively as auxiliary resources for writing enhancement and documentation support.

2. RELATED WORKS

The field of automatic summarization has evolved from unimodal text-based methods to sophisticated multimodal frameworks integrating text, audio, and visual streams. This chapter reviews state-of-the-art (SOTA) literature across these modalities, focusing on their methodologies, strengths, and limitations, to contextualize our behaviour-aware multimodal video summarization framework. A hierarchical taxonomy of summarization approaches is presented in Figure 2, illustrating the interconnections between text, audio, and video summarization, with a focus on behavioural analysis in multimodal contexts.

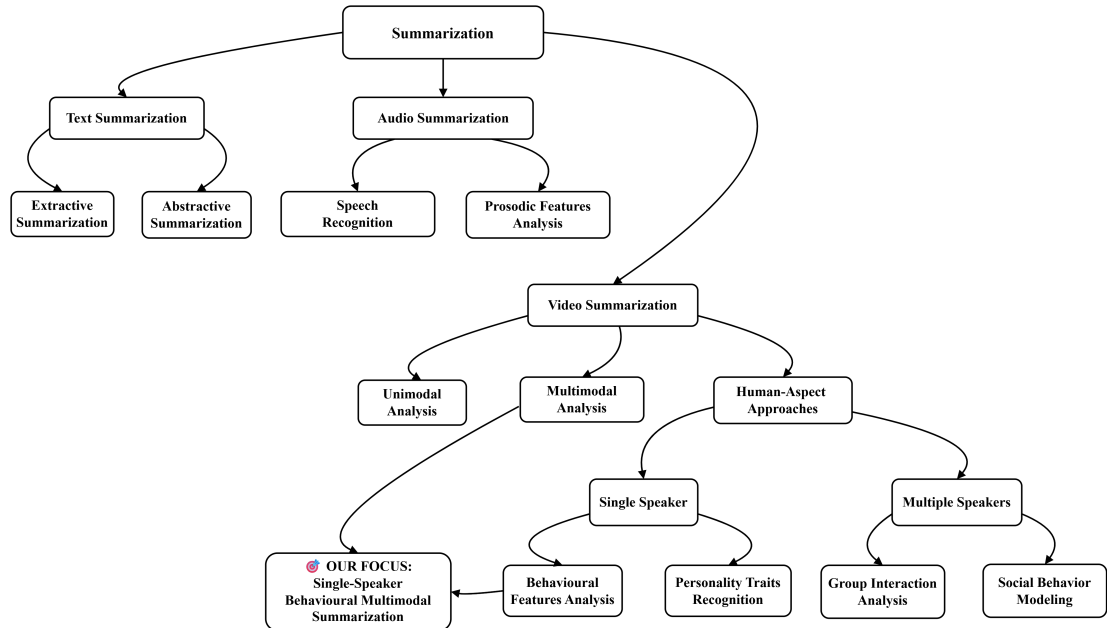


Figure 2. **A hierarchical taxonomy of automatic summarization approaches across text, audio, and video modalities.** The diagram shows the subdivision of video summarization into various analytical approaches, with the research focus highlighted at the intersection of multimodal analysis and single-speaker behavioural feature analysis.

2.1. Text Summarization

Text summarization represents the foundational pillar of automatic summarization research, serving as the basis for more complex multimodal approaches. The field has witnessed substantial evolution from rule-based heuristics to complex neural architectures and LLM-based approaches. Edmundson et al. [27] introduced a seminal framework combining four features, such as word frequency, cue words, title/heading words, and sentence position. This approach outperformed earlier frequency-based methods [28] by integrating linguistic and structural cues, remaining relevant in educational contexts with bonus word and stop word filtering [29]. Statistical methods such as TF-IDF [30] and location-based features [31] further emphasized positional

importance, noting that 85% of key sentences appear at paragraph beginnings [32]. We adopt Edmundson’s *bonus words*-based strategy for our heuristic approach.

The advent of deep learning revolutionized text summarization through both extractive and abstractive paradigms. Transformer-based architectures have achieved remarkable success, with models incorporating local topic information and hierarchical structures showing significant improvements for long document summarization [33]. These approaches address the challenge of capturing both local contextual information and global document structure. Large Language Models (LLMs) have fundamentally transformed summarization capabilities. LLMs have further transformed this field, with models achieving high human preference scores on benchmark datasets [34], highlighting the paradigm shift toward more human-aligned summarization. The integration of sophisticated prompt engineering and instruction tuning has enabled LLMs to perform various summarization tasks without task-specific fine-tuning [35, 36, 37]. This capability has proven particularly valuable for generating pseudo-ground truth summaries in scenarios where human annotations are unavailable, an approach we adopt in our methodology.

2.2. Audio Summarization

Audio summarization has evolved as an important field focusing on extracting key segments and salient events from audio streams to represent the essence of underlying material. The discipline emerged from the fundamental recognition that spoken content required distinct processing approaches, establishing the cascade paradigm that combines automatic speech recognition with text summarization techniques. Early innovations recognized that speech carried prosodic information beyond words, with researchers developing methods to exploit emphasized speech regions using hidden Markov models and non-spectral features to guide content selection [38].

Foundational approaches centered on identifying significant acoustic events that capture human attention through energy and loudness measurement techniques. These methods proved crucial for highlighting audio portions most likely to engage listeners, integrating audio and text modalities to detect conspicuous events that outline main concepts of audio streams [39, 40]. The effectiveness of these approaches became particularly evident in domain-specific applications, where sports broadcasting systems successfully leveraged acoustic event classification to generate targeted highlights, significantly enhancing viewer experience through automated highlight generation [41]. Keyword spotting and voice activity detection emerged as widely adopted methods for audio segmentation and significant event detection, facilitating selection of essential content from continuous streams by detecting specific words or phrases indicating important segments [42]. Researchers transformed one-dimensional speech signals into two-dimensional similarity matrices, enabling detection of recurring themes similar to musical structures, though the effectiveness of these approaches varied significantly across audio classification contexts [43].

The development of neural networks fundamentally transformed audio summarization capabilities through advanced deep learning algorithms that revolutionized audio segmentation. Methods converting audio recordings to text and applying extractive summarization techniques effectively distilled audio conversations,

capturing salient points for meeting and discussion structuring [44]. These systems proved particularly valuable for highlighting key agenda items in professional settings, while complex framework such as the Near-Lossless Semantic Summarization (NLSS) employed multi-dimensional classification to enhance summary quality by categorizing audio units into distinct categories such as silence, music, and speech, compressing non-speech segments accordingly [45]. The emergence of specialized applications has become notable with podcast summarization systems, which addressed unique content discovery challenges where presentation styles and production quality significantly influence user preferences [46, 47]. End-to-end neural systems emerged as significant advances, enabling direct semantic prediction from audio without intermediate token architectures, achieving remarkable accuracy rates across domain prediction, intent recognition, and slot filling tasks [48].

Recent approaches increasingly incorporate contextual information and multimodal integration to enhance summary relevance. The integration of emotional cues, speaker identification, and multimodal features proves important for environments with background noise or multiple speakers. This evolution toward multimodal integration represents an important paradigm shift, recognizing that effective summarization requires understanding beyond speech content alone. Li et al. [49] proposed that language analysis alone is insufficient to capture significant visual and audio events, leading to systems that fuse verbal and non-verbal information through hierarchical attention mechanisms. Existing research addresses persistent challenges in speaker variability, dialectal differences, and the fundamental dependency between recognition accuracy and summarization quality [50]. This field now advances toward foundational models serving multiple downstream tasks including cognitive assessment, speech coding, and content analysis across topological-linguistic approaches spanning monolingual to cross-lingual systems and LLM-based summarization [51]. The emphasis on multimodal integration and adaptive systems capable of handling different formats and genres represents the current frontier, where comprehensive acoustic analysis meets advanced machine learning to create increasingly effective and contextually aware summarization systems.

2.3. Video Summarization

The exponential growth of video content has necessitated intelligent summarization techniques that can distill hours of those video content into meaningful, concise representations. This journey began with simple visual analysis and has evolved toward sophisticated frameworks that understand not just what is shown, but how human communication unfolds through multiple sensory channels. We trace this evolution through three paradigms: unimodal approaches that established foundational principles, multimodal integration that recognized the symphony of information streams, and the emerging frontier of behaviour-aware understanding.

2.3.1. Unimodal Video Summarization

Unimodal video summarization approaches rely on a single information stream, usually visual features, to identify and extract significant content from videos. These methods have evolved significantly over the past two decades, progressing from rule-based heuristics to sophisticated deep learning architectures. Early video summarization methods primarily utilized heuristic and statistical techniques focused on low-level visual features. Initial strategies involved shot boundary detection followed by keyframe extraction to represent video content succinctly [52]. Mundur et al. [53] advanced this by detecting both abrupt and gradual shot transitions using colour histograms and motion, selecting representative frames such as the first, middle, or most representative within each shot. While computationally efficient, these methods struggled with semantic understanding and often produced redundant, visually similar frames [54]. They relied heavily on manually crafted rules, limiting adaptability across diverse video types [55]. Gibson et al. [56] enhanced temporal coherence via sliding window techniques, though at the cost of sensitivity to window parameters. Clustering approaches, such as k-means used by Mundur et al. [53], effectively grouped visually similar frames but faced difficulties with videos exhibiting gradual transitions or complex dynamics [57]. Ejaz et al. [58] introduced improved visual attention frameworks to better capture salient frames. Despite marking a transition towards data-driven methods, these statistical approaches remained constrained by reliance on hand-crafted features, limited semantic capture, and inadequate modeling of complex temporal relationships across video segments [59, 60].

Dictionary-based learning methods sought to identify representative elements, or atoms, that reconstruct videos with minimal error. Cong et al. [61] introduced a sparse coding framework selecting keyframes by reconstruction capability, enhanced later with temporal coherence constraints to improve narrative flow, albeit with added computational complexity. Zhao et al. [62] developed a quasi real-time summarization using online dictionary learning. Gygli et al. [63] formalized this by optimizing a function balancing representativeness, uniformity, and interestingness. Extending this, Sharghi et al. [64, 65] proposed query-focused summarization adapting to user preferences. Despite providing principled optimization frameworks, these methods struggled with computational cost and parameter tuning across varied content [66]. Supervised learning further advanced summarization by leveraging annotated data. Gong et al. [67] pioneered SUMMARIST, utilizing singular value decomposition (SVD) for feature learning, while Potapov et al. [68] employed support vector machines (SVMs) to identify category-specific importance, demonstrating improvements over heuristic baselines.

The integration of deep learning and transformer architectures has transformed video summarization by enabling end-to-end learning directly from raw video data to importance scoring and frame selection. Convolutional neural networks (CNNs) advanced feature representation significantly. Yao et al. [69] proposed a highlight detection framework leveraging CNNs for domain-specific features, while Rochan et al. [70] adapted fully convolutional sequence networks (FCSN) for temporal semantic segmentation. However, CNNs initially lacked robust temporal modeling capabilities, prompting the incorporation of recurrent neural networks (RNNs). Zhang et al. [71]

pioneered the use of LSTMs to capture long-range temporal dependencies, achieving superior performance on benchmark datasets. Ji et al. [72] further enhanced temporal attention via an encoder-decoder LSTM architecture, though challenges persisted with very long videos due to vanishing gradients. Hierarchical LSTM models introduced by Zhao et al. [73, 74] addressed short and long-term dependencies but increased model complexity. Despite improvements, RNN-based methods remain computationally intensive and susceptible to sequential error propagation [75].

Generative adversarial networks (GANs) introduced a new paradigm by framing summarization as adversarial learning. Mahasseni et al. [76] developed an unsupervised adversarial LSTM framework (vsLSTM-GAN), where a generator selects summary frames and a discriminator assesses summary quality. Extensions such as Cycle-SUM [77] and AC-SUM-GAN [78] incorporated cycle consistency and actor-critic reinforcement learning to improve training stability and performance without reliance on annotations. GAN-based methods, however, face challenges including training instability, evaluation difficulties, and limited interpretability [79].

Reinforcement learning (RL) offers a flexible approach to directly optimize summarization objectives. Zhou et al. [4] trained agents balancing representativeness and diversity through reward functions, with Wang et al. [80] extending this via hierarchical rewards capturing local and global video structures. RL methods provide adaptability but often require complex reward design and extensive training for stable performance [81]. Attention mechanisms also enhanced summarization. Fajtl et al. [82] utilized self-attention to model global dependencies, while Muhammad et al. [83] proposed saliency-based frame selection.

Despite these advances, unimodal video summarization approaches suffer inherent limitations. Sole reliance on visual features fails to capture semantic context critical for complex, narrative-driven videos such as interviews or documentaries [9]. They overlook important cues from other modalities, including speech emphasis, background music, or on-screen text [84]. Furthermore, unimodal methods often treat frames independently or with limited temporal context, missing broader narrative arcs or thematic development [85, 86]. Their inability to adapt summaries to user queries or preferences further restricts applicability [64]. These limitations have driven the shift towards multimodal summarization that integrates visual, audio, and textual information to generate richer, and context-aware summaries [2]. Incorporating facial features and emotional cues [87, 88] offers valuable patterns that enhance summarization quality.

Table 1. Summary of State-of-the-art Unimodal Video Summarization Approaches

Authors	Method/Approach	Datasets	Modalities Involved	Key Contributions	Limitations
Mundur et al. [53]	Shot boundary detection	OVP [89]	Visual	Detected shot transitions and keyframe selection	Manual rule dependency
Gibson et al. [56]	Key frame abstraction	Wildlife [56]	Visual	Incorporated temporal context to improve summary coherence	Sensitivity to window size parameters
Ejaz et al. [58]	Visual attention-based keyframe extraction	VSumm [90]	Visual	Improved capture of salient frames via attention mechanisms	Limited semantic understanding; hand-crafted features
Cong et al. [61]	Sparse coding framework for keyframe selection	Kodak [91]	Visual	Balanced reconstruction accuracy and temporal coherence	Computationally expensive; increased complexity
Zhao et al. [62]	LiveLight	Surveillance and YouTube [62]	Visual	Enabled near real-time summarization via online dictionary learning	Parameter tuning issues
Gygli et al. [63]	Objective function combining uniformity, representativeness, and interestingness	SumMe [92], Egocentric [93]	Visual	Introduced supervised learning from human summaries	Requires extensive annotated data
Gong et al. [67]	SUMMARIST	-	Visual	SVD-based keyframe selection from learned visual features	Limited by feature learning capacity
Potapov et al. [68]	Category-specific importance scoring using SVMs	TRECVID [94]	Visual	Improved importance detection by training category-specific classifiers	Relies on category labels; limited scalability
Yao et al. [69]	CNN-based highlight detection framework	YouTube [69]	Visual	CNNs for domain-specific feature learning	Lacked temporal modeling capabilities
Rochan et al. [70]	Convolutional sequence networks adapting semantic segmentation to temporal domain	SumMe [92], TVSum [95]	Visual	Adapted spatial CNN methods to capture temporal video features	Temporal modeling still limited compared to RNNs
Zhang et al. [71]	LSTM-based temporal dependency modeling for keyframe selection	SumMe [92], TVSum [95]	Visual	Pioneered RNN use to model long-range temporal dependencies	Vanishing gradient issues for long videos
Ji et al. [72]	Attentive encoder-decoder LSTM	SumMe [92], TVSum [95]	Visual	Enhanced attention for relevant segment selection in long videos	Difficulty handling very lengthy videos
Zhao et al. [73]	Hierarchical LSTM capturing short and long-term dependencies	SumMe [92], TVSum [95], MED [68], VTW [96]	Visual	Addressed temporal scale issues in video summarization	Increased model complexity and training cost
Mahasseni et al. [76]	vsLSTM-GAN	SumMe [92], TVSum [95], OVP [89], VSumm [54]	Visual	Introduced GAN-based adversarial LSTM framework for unsupervised learning	Training instability; evaluation challenges
Yuan et al. [77]	Cycle-SUM	SumMe [92], TVSum [95]	Visual	Cycle-consistent adversarial LSTM for stable training	Limited interpretability; complex training
Apostolidis et al. [78]	AC-SUM-GAN	SumMe [92], TVSum [95]	Visual	Integrated actor-critic reinforcement learning with GANs	High training complexity
Zhou et al. [4]	Deep reinforcement learning with diversity and representativeness rewards	SumMe [92], TVSum [95]	Visual	Frame selection via reward-based policy learning	Complex reward design; sample inefficiency
Wang et al. [80]	Progressive reinforcement learning with hierarchical rewards	SumMe [92], TVSum [95]	Visual	Captured both local and global video structure in rewards	Increased training complexity and resource needs
Fajtl et al. [82]	Self-attention mechanisms to capture global temporal dependencies	SumMe [92], TVSum [95]	Visual	Enhanced temporal dependency modeling via attention	Computationally intensive

2.3.2. Multimodal Video Summarization

The primary objective of multimodal video summarization is to produce concise and informative representations that incorporate various data modalities such as visual information (video frames), auditory information (audio tracks), and textual information (transcripts and descriptions) [97]. Early efforts by Smith et al. [98] combined visual analysis with speech recognition, aligning these modalities temporally to detect segments of joint significance. Ma et al. [99] advanced this by explicitly modeling cross-modal attention to improve identification of perceptually important frames beyond visual cues alone. Evangelopoulos et al. [84] developed an audio-visual-text saliency framework that generated modality-specific saliency curves fused through weighted combination. Raventos et al. [100] applied dynamic time warping to synchronize audio-visual features with commentary in sports videos, achieving precise temporal alignment crucial for fast content. Gygli et al. [92] introduced supervised models trained on human summaries, demonstrating the value of learning from preferences despite the expense of annotation. These approaches significantly outperformed unimodal methods but often depended on simple fusion strategies and rule-based modality integration, which limited flexibility. Moreover, temporal misalignment across modalities remained a challenge, affecting the coherence and accuracy of generated summaries [101, 102].

The emergence of deep learning architectures revolutionized multimodal summarization by enabling joint representation learning across modalities. Zhu et al. [103] introduced a topic-aware multimodal attention network that utilized separate visual and audio encoders combined with a cross-modal attention mechanism to dynamically weight relevant features for summary generation. Narasimhan et al. [23] furthered this direction with their "TL;DW" framework for instructional videos, aligning textual task descriptions with visual content in a joint embedding space to facilitate query-focused summarization. Park et al. [8] proposed the Multimodal Frame-Scoring Transformer (MFST), which employed self-attention to model temporal dependencies explicitly across visual, audio, and textual modalities, achieving strong benchmark performance. Zhao et al. [104] presented a hierarchical multimodal transformer employing a two-tier attention mechanism, within and across modalities, preserving modality-specific characteristics while capturing cross-modal correlations. Guo et al. [105] addressed modality misalignment in CFSum through coarse-to-fine temporal fusion using multi-head attention, proving effective for asynchronous content such as voice-overs. Cross-modal attention has become central for dynamic fusion across modalities, with fusion strategies broadly categorized as early (feature-level), late (decision-level), and hybrid approaches. Wei et al. [106] employed early fusion by concatenating multimodal features before deep learning, though faced challenges in modality alignment and scale differences. Li et al. [107] developed a hybrid fusion method combining separate and joint modality processing with attention to optimally weight features, excelling with asynchronously aligned videos. Lin et al. [5] introduced VideoXum, which used bidirectional attention between visual and textual streams bridged by audio, effectively handling semantic alignment despite temporal shifts.

The introduction of transformer architectures marked a significant advancement in multimodal integration. These architectures have become foundational in these

developments, enabling joint representation learning and dynamic fusion of visual, audio, and textual data. Palaskar et al. [12] proposed a multi-source sequence-to-sequence model with hierarchical attention that integrates information from video and audio transcripts to generate fluent textual summaries of instructional videos, moving beyond traditional text compression to fuse information from different modalities into coherent abstractive summaries. Hierarchical models, such as the MAST framework [108], employed trimodal hierarchical attention to prioritize text while effectively integrating audio and video. The power of transformer-based models lies in their ability to capture both local and global dependencies, a key factor for generating coherent and contextually rich summaries [104, 86]. Memory-augmented architectures further improve the modeling of long-range temporal dependencies across modalities. Feng et al. [109] proposed extractive summarization enhanced by memory-augmented neural networks to capture long-range multimodal dependencies. Building upon this, He et al. [110] developed MA-LMM, a memory-augmented large multimodal model that maintains separate modality-specific memories coordinated via a cross-modal controller, achieving notable improvements in summarizing lengthy videos.

Recent advances in large language models (LLMs) have significantly advanced multimodal video summarization by enabling deeper semantic understanding and more effective cross-modal reasoning. Vision-language pretraining models such as CLIP have enhanced feature representation, exemplified by CLIP-It [7], which aligns video frames with textual queries in a shared embedding space to enable adaptive, and query-focused summarization. Similarly, Sun et al. [111] introduced VideoBERT, a joint video-language model pretrained on vast video-text datasets, improving temporal segmentation and importance scoring by capturing rich cross-modal relationships. Further leveraging LLMs, Argaw et al. [6] proposed a framework that generates pseudo-labels for weakly supervised training, overcoming data scarcity and achieving state-of-the-art results by reasoning across modalities at multiple temporal scales. Lynch et al. [112] utilized large vision-language models (VLMs) to create abstractive summaries that integrate visual frames and transcripts, substantially improving semantic coherence. Complementing these approaches, prompt engineering techniques have gained prominence, such as Hua et al. [20] fine-tuned LLMs using temporal prompt instruction tuning to preserve narrative flow across modalities, while Zhang et al. [113] enhanced extractive summarization with diffusion models guided by LLM reasoning, illustrating the synergy between generative models and traditional summarization.

The advancement of multimodal summarization models is strongly supported by comprehensive datasets, such as TVSum [95], SumMe [92], MM-AVS [97], among others, which facilitate learning intricate cross-modal relationships. Such datasets also help mitigate modality bias, where models might overemphasize one modality at the expense of others [114, 115]. Hybrid fusion methods, in particular, balance joint and separate processing streams using attention mechanisms, enabling effective handling of asynchronous modalities such as background music or voice-overs. Complementing these advancements, contrastive learning techniques such as CLIP [15] have also been adopted to learn robust cross-modal alignments without the need for extensive supervised data, contributing to improved generalization and performance in multimodal tasks.

Despite significant progress, multimodal video summarization faces ongoing challenges. Many models exhibit limited generalizability across different content genres, highlighting the need for more robust transfer learning techniques [116, 85]. The computational demands of large vision-language models (VLMs) hinder deployment in real-time and resource-limited environments. Furthermore, most approaches focus on general content importance and lack fine-grained personalization capabilities that adapt to user-specific contexts [117, 118].

Table 2. Summary of State-of-the-art Multimodal Video Summarization Approaches

Authors	Method/Approach	Datasets	Modalities Involved	Key Contributions	Limitations
Smith et al. [98]	Integration of language and image understanding techniques	-	Visual, Text	High compression ratios (20:1); alignment of visual and textual saliency	Limited fusion methods; scalability constraints
Ma et al. [99]	User attention model integrating visual and audio features	-	Visual, Audio	Modeled cross-modal attention to improve content detection	Fusion methods not effective
Evangelopoulos et al. [84]	Multimodal saliency fusion via weighted linear combination	MovSum [84]	Visual, Audio, Text	Multi-saliency curve fusion for unified importance scoring	Relied on simple fusion; limited adaptability
Raventos et al. [100]	Synchronization of audio-visual descriptors with commentary using dynamic time warping	Sports [100]	Visual, Audio	Achieved temporal precision critical for fast-paced events	Domain-specific; limited generalizability
Otani et al. [119]	DNN for joint video-description semantic mapping and clustering-based summarization	SumMe [92]	Visual, Text	Deep semantic features encoding objects, actions, and scenes	Dependency on paired video-description data
Sharghi et al. [64]	Query-focused video summarization using probabilistic model	Egocentric [93], VideoSet [120]	Visual, Text	Enabled user-adaptive summaries with principled optimization	Complex optimization; tuning difficulties
Zhu et al. [103]	Topic-aware multimodal attention network with separate encoders	Instructional videos	Visual, Audio	Dynamic feature weighting across modalities	Requires domain-specific data
Narasimhan et al. [7]	CLIP-It	SumMe [92], TVSum [95], QFVS [65]	Visual, Text	Language-guided transformer for generic and query-focused video summarization	Dependency on dense video captioning; limited query diversity evaluation
Narasimhan et al. [23]	TL:DW	WikiHow, CrossTask [121], COIN [122]	Visual, Text	Query-focused summarization adapting to user interest	Complex joint embedding training
Park et al. [8]	Multimodal Frame-Scoring Transformer (MFST)	SumMe [92], TVSum [95]	Visual, Audio, Text	Modeled temporal dependencies across modalities via self-attention	Modality misalignment challenges
Zhao et al. [104]	Hierarchical multimodal transformer with two-tier attention	SumMe [92], TVSum [95]	Visual, Audio, Text	Captured local and global cross-modal dependencies	Increased model complexity
Guo et al. [105]	Transformer-based coarse-fine fusion for modality alignment	TVSum [95], YouTube [123], QVHighlights [124]	Visual, Audio, Text	Addressed asynchronous modality fusion challenges	Model complexity and scalability
Wei et al. [106]	SASUM	SumMe [92], TVSum [95]	Visual, Audio, Text	Early fusion by concatenating multimodal features before deep learning	Modality scale differences and alignment issues
Lin et al. [5]	VideoXum	VideoXum [5], SumMe [92], TVSum [95]	Visual, Audio, Text	Bidirectional attention between visual and textual streams bridged by audio	Requires modality balancing
Feng et al. [109]	Extractive summarization with memory-augmented neural networks	SumMe [92], TVSum [95]	Visual, Audio, Text	Captured long-range multimodal dependencies	Increased model size and complexity
He et al. [110]	MA-LMM	COIN [122], LVU [125], Breakfast [126]	Visual, Audio, Text	Modality-specific memory-augmented large multimodal summarization	High computational cost
Argaw et al. [6]	Transformer-based encoder-decoder with LLM-generated pseudo ground-truth summaries	LfVS [6], SumMe [92], TVSum [95]	Visual, Text	Large pretraining dataset; autoregressive approach addressing class imbalance	Dependency on speech-video alignment; potential LLM bias in dataset generation
Sun et al. [111]	VideoBERT	YouTube, YouCook II [127]	Visual, Audio, Text	Visual-linguistic model with BERT over vector-quantized tokens	Loss of fine-grained visual information through quantization

2.3.3. *Behavioural Approaches in Video Summarization*

Video summarization inherently intersects with human and behavioural aspects due to its reliance on interpreting visual cues that are rich in social and emotional content. Behavioural research in video summarization has demonstrated that non-verbal cues and contextual elements significantly influence audience engagement, decision-making, and emotional responses [128, 129].

Recent research has increasingly emphasized the critical role of behavioral dimensions in multimodal video summarization, recognizing that communicative and physiological cues provide essential context for identifying truly significant content. A notable example is the work by Ma et al. [130], who developed a human-machine collaborative summarization framework integrating real-time physiological measurements, such as pupillary responses, to capture viewer engagement and attention. By correlating these physiological signals with traditional audio-visual features, their approach enhances importance estimation, producing summaries that better align with perceptual relevance. This innovative use of pupillary response introduces a novel behavioral index, but not yet exploited in traditional summarization techniques which primarily focus on content features.

Complementing physiological metrics, behavioural influences on video consumption have been explored in the context of video platforms, where visual cues significantly affect user decision-making, especially among college demographics [131, 132]. The addictive characteristics of such platforms, as described by Liao [132], reveal how video content impacts cognitive processing, underscoring the need for summarization frameworks that not only condense content but also cater to behavioural patterns to improve viewer satisfaction. Rochan et al. [133] highlight the importance of supervised learning methods that align content representation with viewer cognitive and psychological responses. Saquil et al. [117] further advanced personalization in video summarization by employing multiple pairwise ranking networks to model diverse user preferences across primarily visual features, while proposing extensions to incorporate auxiliary multimodal information.

Despite these advances, explicit modeling of multimodal behavioural features, such as gestures, facial expressions, and vocal prosody, remains limited, even though these are fundamental to human communication and content interpretation. Systematic integration of such behavioural features into multimodal summarization frameworks remains an open and vital challenge. Addressing this gap is essential to developing human-centric summarization methods that faithfully represent the nuanced interplay of verbal and non-verbal communication, ultimately enhancing user engagement and experience.

2.4. Gaps in Existing Research

Our behaviour-aware multimodal video summarization framework introduces several distinctive strategies that differentiate from existing approaches in the literature. The following analysis highlights these key differences:

Explicit Behavioural Features Modeling. Unlike most existing approaches that focus primarily on content-based features, our both proposed frameworks explicitly model behavioural cues across modalities as indicators of importance. While Ji et al. [72, 86] incorporate attention mechanisms for multimodal integration, they don’t specifically target behavioural signals such as gestures, facial expressions, or vocal modulations that indicate emphasis. Zhao et al.’s [104] hierarchical transformer approach processes visual and audio features, but lacks explicit behavioural modeling that recognizes communicative signals such as head movements or emotional transitions. In contrast, our approach extracts and analyzes specific behavioural features such as head movements tracked via MediaPipe Pose [13], emotional transitions detected through DeepFace [14], and prosodic patterns captured through YAAPT [16] and OpenSMILE’s eGeMAPS features [17].

Temporal Alignment. While many approaches acknowledge the importance of temporal alignment, our both proposed frameworks implement millisecond-precision synchronization. Most existing methods [60, 22, 20] use coarse alignment between modalities, often at the segment or shot level. Our approach employs the Montreal Forced Aligner (MFA) [134] to achieve precise word-level timestamp mapping (with millisecond accuracy), enabling detection of micro-level correspondences between spoken words and accompanying visual or auditory emphasis.

Feature Extraction Approach. Our transformer-based framework employs a more comprehensive feature extraction approach across modalities. We combine state-of-the-art models for each modality: CLIP [15] for visual embeddings, HuBERT [18] for audio processing, and RoBERTa [19] for textual analysis. These are enhanced with specialized behavioural features such as facial movement trajectories, emotional transitions, pitch contours, voice quality metrics, and loudness patterns.

Multimodal *Bonus Words* Identification. Our heuristic framework introduces the concept of *bonus words*, following Edmundson’s principles [27], as multimodal anchors for summarization purpose. This approach uniquely identifies textual terms that receive behavioural emphasis (through gestures, vocal stress, or facial expressions), using these *bonus words* as weighted anchors for summary generation. This creates a direct bridge between behavioural signals and semantic content.

Adaptive Threshold Mechanism. Our heuristic framework introduces an adaptive threshold system for identifying significant behavioural cues. We implement a threshold mechanism that dynamically adjusts based on the statistical distribution of behavioural features within each video. This dynamic approach allows for context-sensitive detection of significant cues.

Modality Integration Strategy. Our transformer-based cross-modal integration approach fundamentally differs from existing fusion methodologies in video summarization. While conventional approaches process modalities independently before applying late-stage fusion techniques [84, 7, 135, 105], this framework implements a unified cross-modal attention mechanism that enables dynamic inter-modal conditioning throughout the feature learning process. This encoder-decoder

architecture with integrated cross-modal attention facilitates direct learning of complex behavioural correlations across modalities, contrasting with hierarchical approaches [73, 74, 104] that maintain distinct processing pipelines for each modality before convergence. By allowing modalities to mutually influence each other’s representations from the earliest stages of processing, our approach captures richer semantic relationships and produces more coherent multimodal understanding.

Dataset Focus and Evaluation Methodology. We make distinct choices in dataset selection and evaluation strategy. While most approaches evaluate on general datasets such as TVSum [95] and SumMe [92] that focus primarily on action-oriented content, we specifically target the ChaLearn First Impressions dataset [136] with its rich behavioural content in controlled interview settings. Our evaluation methodology combines traditional metrics with behaviour-specific assessment, using LLM-generated pseudo-ground truth summaries following previous state-of-the-art approaches [23, 6].

Our behaviour-aware multimodal video summarization framework represents a significant advancement over existing approaches by explicitly modeling the behavioural dimensions of communication that are crucial for human understanding but typically overlooked in content-focused methods. By detecting and integrating communicative cues across visual, audio, and textual modalities, we create summaries that preserve not just what was said, but how it was expressed. This approach is particularly valuable for human-centric content such as interviews, presentations, and instructional videos, where behavioural signals carry significant semantic weight.

3. METHODOLOGY

This chapter outlines the proposed methodology for our behaviour-aware multimodal video summarization approach, which overcomes the limitations of traditional methods by integrating visual, audio, and textual data to capture nuanced behavioural cues. The proposed methodology employs a dual-pipeline strategy, (1) a heuristic-based approach and (2) a transformer-based approach, both supported by shared preprocessing and feature extraction components. This methodology ensures temporal alignment across modalities and prioritizes behaviourally significant patterns to generate coherent and contextually relevant video summaries.

Figure 3 illustrates the overall architecture of our methodology, consisting of five interconnected components: data sources, preprocessing pipeline, multimodal feature extraction, dual summarization frameworks, and finally, evaluation through pseudo-ground truth summary generation for evaluation. Each component has been designed to maintain temporal alignment across modalities while extracting behaviourally significant patterns that contribute to more meaningful video summaries.

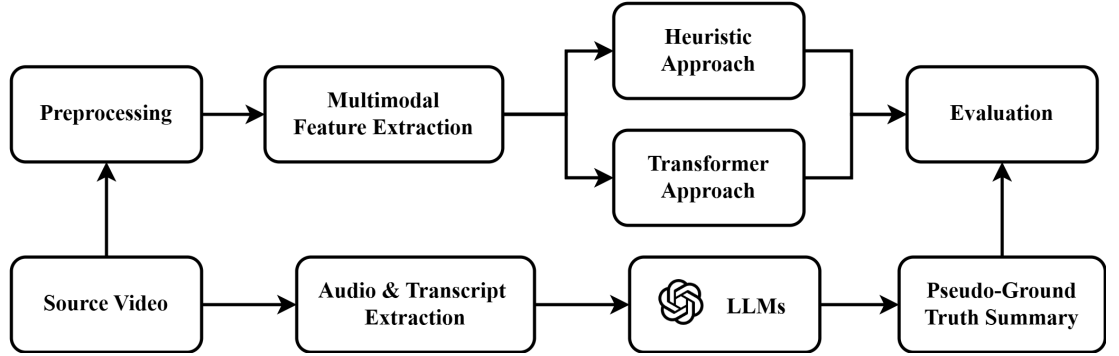


Figure 3. **Systematic architecture for behaviour-aware multimodal video summarization.** This diagram presents the overall system workflow, illustrating the dual approach to video summarization.

3.1. Overview of the Methodology

Our methodology leverages the ChaLearn First Impressions dataset [136], selected for its rich behavioural content in interview-style videos, and ideal for analyzing multimodal cues. The preprocessing pipeline transforms raw videos into synchronized streams of visual frames, audio segments, and timestamped transcripts, with particular emphasis on millisecond-precision alignment between modalities using the Montreal Forced Aligner (MFA) [134]. This temporal coherence is essential for accurately correlating behavioural cues across different channels. A distinctive aspect of our methodology is the implementation of two summarization frameworks:

1. **Framework I - Heuristic approach:** This interpretable approach identifies key words, termed as *bonus words* that receive emphasis across multiple modalities, using them to weight sentence importance for summary inclusion.

2. **Framework II - Transformer-based approach with cross-modal attention:**

This approach employs transformer-based encoders for each modality, integrated through cross-modal attention, with an autoregressive decoder generating temporally coherent summaries.

For visual feature extraction, we implement a comprehensive approach to capture both explicit behavioural signals and semantic visual content. We detect facial movements and posture shifts using MediaPipe Pose [13], which provides precise tracking of key facial landmarks to identify significant head movements, gestures, and non-verbal cues. Emotional state analysis is performed using DeepFace [14], enabling the detection of transitions between emotional states such as neutral to happy or surprised, which often indicate narrative significance. For Framework II specifically, these traditional behavioural features are complemented by deep semantic visual embeddings from the CLIP model [15], which uses contrastive learning from 400 million image-text pairs to capture rich visual context beyond explicit behavioural cues.

In the audio domain, we extract a range of prosodic and spectral features to capture the expressive elements of speech that often convey behavioural intent. Pitch contours are extracted using YAAPT [16], which employs a hybrid approach combining spectral and temporal analyses to robustly track fundamental frequency even in challenging audio conditions. Voice quality and loudness are quantified using the openSMILE toolkit [17], with particular focus on the Hammarberg Index for spectral balance and normalized energy measures for perceived intensity. In Framework II, these traditional acoustic features are enhanced with contextual audio representations from the HuBERT model [18], which captures the rich temporal dynamics of speech through self-supervised learning, providing embeddings that reflect both linguistic content and paralinguistic cues.

For textual feature analysis, we implement multiple complementary approaches to identify both the semantic significance and the structural importance of words within the transcript. We apply TF-IDF analysis [30] to highlight content-rich terms that distinguish specific segments from general discourse. Named entity recognition [137] is used to identify references to people, organizations, or concepts that often represent anchor points in the narrative. In Framework II, we employ contextual textual embeddings using RoBERTa [19], which provides deep semantic representations that capture the nuanced meaning of words in context, including sentiment, emphasis, and rhetorical structure. This multi-layered approach to textual analysis ensures that both frameworks can effectively identify significant content, with Framework I relying on statistical features and Framework II leveraging both statistical and transformer-based approaches.

For evaluation, we employ large language models (LLMs) to generate pseudo-ground truth reference summaries against which both frameworks are assessed using comprehensive text-based and video-based evaluation metrics. Our methodology is guided by several key design principles addressing fundamental challenges in behaviour-aware multimodal video summarization:

1. We emphasize explicit modeling of behavioural cues such as gestures, facial expressions, and vocal intonations that convey communicative intent [23, 112].

2. We prioritize the identification and selection of moments where multiple modalities exhibit aligned patterns of emphasis or expressiveness [104, 138].
3. We maintain temporal coherence through preservation of chronological structure and narrative flow in generated summaries [103].
4. We ensure adaptability to context through sensitivity to the interview setting, prioritizing features relevant to this genre of communication [130, 99].
5. We implement complementary frameworks, combining both heuristic and transformer-based approaches to provide comprehensive understanding and accommodate different requirements for interpretability and computational complexity [8, 139].

The rest of the chapter details each component of our methodology, starting with data sources, followed by preprocessing, multimodal feature extraction, the development of our two summarization frameworks, and finally, our approach to evaluation through pseudo-ground truth summary generation using LLMs.

3.2. Data Sources

We utilize the ChaLearn First Impressions dataset [136] for our multimodal video summarization framework, a comprehensive and publicly available resource specifically designed for analyzing human behaviour in video content. This dataset comprises 10,000 high-quality video clips, each ranging from 8 to 20 seconds in duration, featuring 7,138 unique subjects engaged in short job interview tasks. While the dataset was originally curated for personality trait recognition and interview score assessment, we find its rich multimodal content makes it an ideal choice for video summarization research. We value the clips’ diverse range of speakers, varying in age, gender, ethnicity, and linguistic backgrounds, as this ensures broad representativeness and generalizability of our proposed framework.

3.2.1. Dataset Characteristics

We observe that each video clip in the ChaLearn First Impressions dataset is recorded at an average of approximately 24 frames per second (fps), providing sufficient temporal resolution to capture subtle behavioural cues such as facial expressions, head movements, and gestures. We appreciate that the audio is recorded in a controlled environment, ensuring clear speech with minimal background noise, which is important for our accurate speech transcription and prosodic feature extraction. We note that on average, each clip contains a transcript of approximately 38 words (152 characters), reflecting concise yet information-rich spoken content typical of interview scenarios. The standardized format of the dataset, with consistent video resolution and audio quality (16 kHz sampling rate), allows us to implement uniform processing across all clips, reducing preprocessing variability in our pipeline.

We identify the dataset’s behavioural richness as a key asset for our multimodal analysis. We observe that speakers exhibit a wide range of non-verbal cues, including

facial expressions (e.g., smiles, frowns), head tilts, hand gestures, and body posture shifts, which convey emotional and communicative intent. We also find that verbal content includes varied speech patterns, such as changes in pitch, loudness, and pauses, reflecting prosodic emphasis. We recognize that the diversity in accents and speaking styles further enhances the dataset’s suitability for testing the robustness of our summarization framework across different linguistic and cultural contexts.

To balance computational feasibility with the need for comprehensive evaluation, we select a subset of 1,500 video clips for our study. We design our selection process to maintain the dataset’s diversity while ensuring manageable processing demands for deep learning models. We employ a stratified sampling approach, where we choose clips to preserve the distribution of demographic attributes (e.g., gender, age) and behavioural characteristics (e.g., expressiveness, gesture frequency). This subset includes approximately 1,071 unique subjects, ensuring we have a broad representation of individual differences in communication styles. We find that the selected clips collectively cover a wide range of interview topics, from self-introductions to responses about professional experiences, providing varied semantic content for our summarization tasks.

3.2.2. *Rationale*

We choose the ChaLearn First Impressions dataset over other video datasets (e.g., TVSum [95], SumMe [92]) due to its focus on human-centric communication, which aligns with our framework’s emphasis on behavioural and prosodic cues. Unlike datasets with diverse genres such as, movies [84, 55], and vlogs [138], we find that the controlled interview setting provides a consistent context for evaluating summarization performance, reducing confounding factors such as varying video styles or complex scene transitions. We recognize that the dataset’s short clip duration also mirrors real-world applications, such as summarizing brief professional or educational videos, enhancing the practical relevance of our framework. We identify several key features that make the ChaLearn First Impressions dataset particularly well-suited for our multimodal video summarization research:

1. We leverage the fact that each clip integrates visual (facial expressions, gestures), audio (speech, prosody), and textual (transcribed content) modalities, enabling both of our frameworks to capture a holistic representation of human communication. This aligns with our goal of integrating semantic and behavioural cues for summarization.
2. We take advantage of the short duration of clips (8-20 seconds) and their precise timestamps, which allow for the fine-grained alignment of modalities important for generating our timestamp-aligned summaries. We benefit from the controlled setting that ensures audio and video streams are synchronized, facilitating accurate cross-modal analysis.
3. We value how the dataset captures naturalistic behaviours in a semi-structured context (job interviews), where speakers exhibit spontaneous communication. This variability supports our evaluation of the framework’s ability to detect

emotionally and semantically significant moments, such as emphasized words or expressive gestures.

4. We appreciate the consistent video and audio quality which simplifies our preprocessing and feature extraction, ensuring that differences in summarization performance are attributable to our framework rather than data inconsistencies.

3.2.3. Data Accessibility and Ethical Considerations

We access the ChaLearn First Impressions dataset through the ChaLearn Looking at People challenge platform¹, ensuring reproducibility and transparency in our research. We verify that the dataset complies with ethical guidelines, with all participants providing informed consent for the use of their video and audio data in research. To further ensure ethical usage, we anonymize the selected subset, and we do not use any personally identifiable information in our analysis. We adhere to data privacy standards throughout our experiments, processing only the provided audio-visual content and transcripts without additional metadata that could compromise participant anonymity.

We also consider the ethical implications of behavior-aware video summarization, particularly regarding potential biases in the dataset or in our summarization algorithms. We acknowledge that interview scenarios might introduce certain cultural or demographic biases, and we strive to mitigate these effects through our diverse sampling approach. We design our feature extraction and summarization methods to focus on behavioural significance rather than subjective judgments about the speakers, aiming for a fair representation that preserves the communicative intent across different demographic groups. Throughout our research, we maintain transparency about the capabilities and limitations of our framework, especially when it comes to interpreting complex human behaviors across diverse populations.

3.3. Preprocessing Pipeline

The preprocessing phase is an important foundation for the multimodal video summarization framework, ensuring that raw video data is transformed into structured, synchronized streams of visual, audio, and textual information. This stage involves extracting parallel data streams from the input videos, standardizing their formats, and aligning them temporally to enable seamless integration during feature extraction and summarization.

3.3.1. Frame Extraction

We implement a systematic frame extraction process as a foundational step in our multimodal video summarization framework, designed to convert the visual content of video clips into a structured sequence of images suitable for subsequent analysis. We apply this process to the ChaLearn First Impressions dataset, consisting of 1,500

¹ChaLearn First Impression Dataset

short video clips ranging from 8 to 20 seconds in duration, recorded at an average of approximately 24 frames per second (fps). Our primary objective is to extract a representative subset of frames that captures essential visual information, such as facial expressions, head movements, and gestures, which are important for identifying behavioural cues in interview settings.

We strategically sample frames at a rate of 1 frame per second, balancing computational efficiency with the need to preserve temporal detail necessary for effective summarization. This approach is based on our observation that behavioural signals, such as a speaker’s smile, head tilt, or hand gesture, typically evolve over intervals of several hundred milliseconds to a few seconds, making it unnecessary to process every frame of the original video. We significantly reduce the processing load through this sampling method, for example, a 20-second clip is reduced from 480 frames to approximately 20 frames, while ensuring that key moments of expressive significance are retained.

We begin our extraction process by loading each video clip into memory using OpenCV [140], ensuring that the video’s visual content is accessible for sampling with timestamps calculated to maintain alignment with other modalities. We sample frames at uniform intervals, for example, selecting every 24th frame for a 24 fps video, corresponding to a 1-second interval. This uniform sampling ensures even distribution of extracted frames across the clip’s duration, providing comprehensive coverage of the visual narrative. We assign each sampled frame a precise timestamp derived from its frame index divided by the video’s frame rate, which is crucial for synchronizing visual data with audio and textual information during cross-modal analysis.

We store the extracted frames as individual PNG files, preserving the native resolution of the ChaLearn dataset. We select the PNG format for its lossless compression, which maintains the visual quality of the frames and ensures that subtle details like facial expressions or gestures remain intact. We assign each frame a unique identifier linking it to its source video, timestamp, and frame index, organizing them in a hierarchical directory structure with subfolders for each video clip to manage the approximately 30,000 frames generated across the dataset. This structured approach facilitates efficient retrieval during subsequent processing stages and ensures robust traceability throughout the pipeline.

3.3.2. Audio Processing

We implement a robust audio extraction process as a key preprocessing step in our multimodal framework, focused on isolating clean audio streams from video clips to enable accurate speech transcription and prosodic feature extraction. We leverage FFmpeg [141], a versatile multimedia processing toolkit, to separate and standardize the audio into a format optimized for downstream tasks. Our extraction process specifies the output as waveform audio file format (.wav) with a 16 kHz sampling rate.

We select this 16 kHz sampling rate because it effectively captures the full human speech spectrum (100 Hz to 8 kHz) while minimizing data size, aligning with the input requirements of our speech processing models, including Whisper [142] for transcription and HuBERT [18] for audio embeddings. We configure the extraction to

produce single-channel (mono) output, which eliminates potential stereo discrepancies and ensures consistent processing across all videos in the ChaLearn dataset.

We maintain precise temporal synchronization with the original video during extraction, preserving exact audio duration with timestamps starting at 0 seconds. This temporal fidelity is essential for subsequent cross-modal integration, ensuring that audio cues can be accurately aligned with visual and textual elements. The output of this process is a collection of high-quality WAV files that serve as the foundation for both transcription services and audio feature extraction, establishing a clean baseline for prosodic and spectral analysis in our summarization framework.

3.3.3. *Speech-To-Text Conversion*

We develop a comprehensive speech-to-text conversion pipeline to generate accurate textual transcripts from the extracted audio, forming the basis for our textual modality analysis. We employ OpenAI’s Whisper model [142] for automatic speech recognition (ASR), as illustrated in Figure 4. We select this model for its exceptional performance across diverse accents and speaking styles, and its resilience to minor background noise that may be present in the ChaLearn dataset. These characteristics make it particularly well-suited for accurately transcribing the varied speech patterns found in interview scenarios.

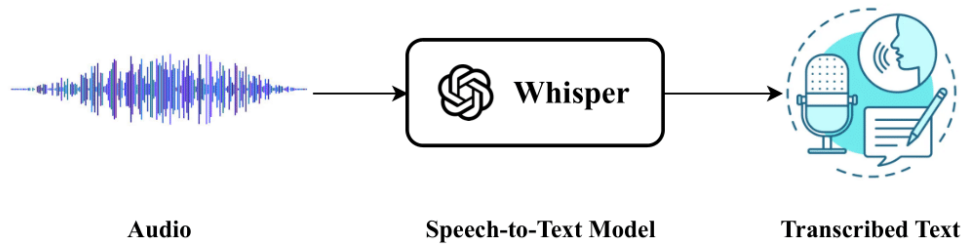


Figure 4. **Speech-to-Text generation process using OpenAI’s Whisper API [142].**

We configure Whisper with English language detection and process each audio file to produce raw transcripts that are naturally segmented into sentences or phrases based on pauses and intonation patterns in speech. To enhance the structural coherence of these transcripts, we implement a refinement step using SpaCy [143], which performs sentence boundary detection to correct run-on sentences, adds appropriate punctuation, and normalizes text by converting to lowercase and removing extraneous whitespace.

We store the refined transcripts in JSON format, with each sentence or phrase initially linked to approximate timestamps derived from Whisper’s output. These timestamps provide a preliminary temporal anchoring that is later refined through our forced alignment process to achieve precise synchronization. This refinement step ensures that our transcripts are grammatically coherent and properly structured for subsequent feature extraction, sentiment analysis, and summarization tasks. The output is a set of clean, well-structured transcripts that accurately represent the spoken content while preserving its temporal relationship to other modalities.

3.3.4. Timestamp Alignment

We implement a precise temporal alignment process to synchronize transcribed text with corresponding audio and video segments, enabling accurate cross-modal analysis. This step ensures that behavioural cues detected in visual frames can be correctly associated with spoken content and vocal prosody. We employ the Montreal Forced Aligner (MFA) [134] to assign millisecond-accurate timestamps to each word in the transcript, as illustrated in Figure 5.

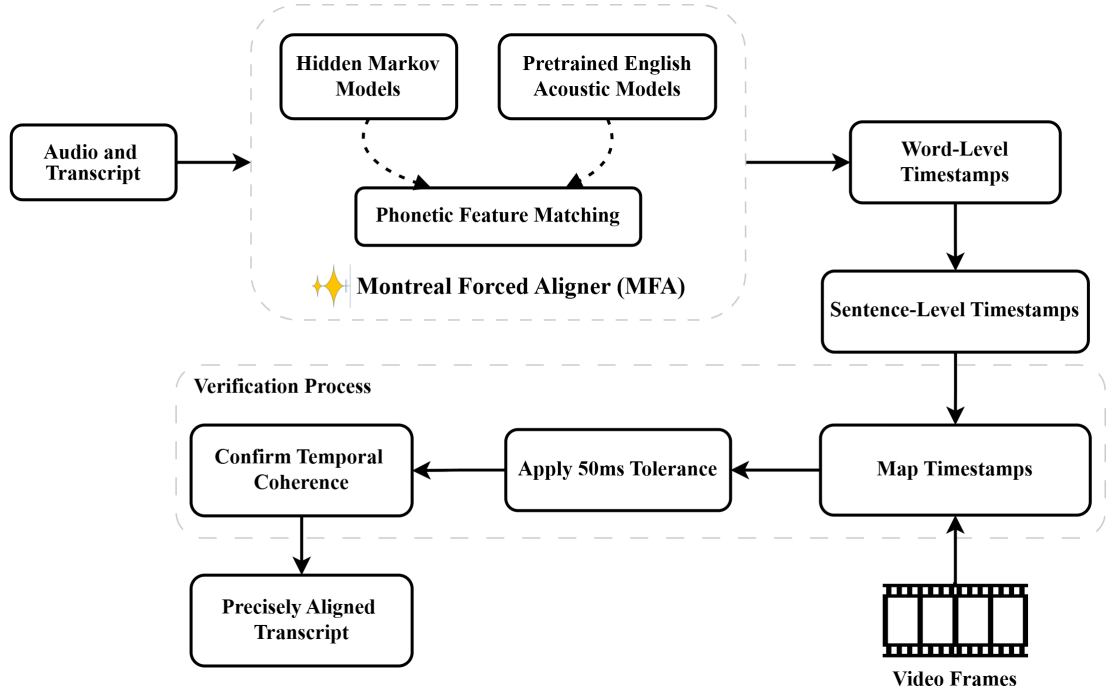


Figure 5. **Montreal Forced Aligner (MFA) timestamp alignment process.** This diagram illustrates the end-to-end process of synchronizing transcribed text with audio and video streams.

We evaluate multiple alignment tools, such as Gentle², Aeneas³, Google Cloud’s speech-to-text API⁴ or Whisper’s [142] built-in alignment, before selecting MFA for its superior precision and robustness. MFA leverages hidden Markov models and Kaldi-based pretrained English acoustic models to match the audio’s phonetic features to the text, generating start and end timestamps for each word. This approach provides word-level alignment precision averaging $\pm 10\text{ms}$, substantially more accurate than alternatives such as Gentle, Aeneas, or Whisper’s built-in alignment capabilities. We find that MFA handles diverse speech rates and pronunciations effectively, ensuring reliability across the varied speakers in our dataset.

Our alignment process begins by inputting the refined transcript (as plain text) and the WAV audio file into MFA. After processing, we derive sentence-level timestamps by aggregating word timestamps, for instance, a sentence’s start time corresponds

²Gentle Forced Aligner

³Aeneas Forced Aligner

⁴Google Cloud’s Speech-to-Text

to its first word’s start time, and its end time to its last word’s end time. We save the results as TextGrid files and convert them to JSON format for integration with our processing pipeline. We verify alignment accuracy to ensure sentence timestamps correspond to video frames (e.g., 7.21s to 8.51s maps to frames 173-204 at 24 fps), with a 50ms tolerance for minor discrepancies. We find MFA’s open-source nature and customizable acoustic models particularly valuable, as they allow adaptation to specific dataset characteristics and support future scalability. The resulting aligned transcripts provide a robust foundation for cross-modal feature extraction and integration, enabling our framework to detect moments where visual, audio, and textual cues coincide to signal behavioural or semantic significance.

3.4. Multimodal Feature Extraction

We develop a comprehensive multimodal feature extraction methodology that serves as the analytical foundation of our video summarization framework. This process involves identifying behaviourally and semantically significant patterns across visual, audio, and textual modalities to detect key moments for inclusion in the summary. Our approach is designed to capture the rich interplay of behavioural cues that typically accompanies important content in human communication, particularly in interview settings where non-verbal signals often reinforce verbal content.

We implement two distinct approaches to feature extraction that share common preprocessing components but diverge in their analytical methodology. Framework I employs a transparent, rule-based approach focused on directly observable behavioural cues, while Framework II enhances these with contextual embeddings to capture richer semantic representations. This dual approach enables us to compare the efficacy of interpretable heuristic methods against more complex neural representations for behaviour-aware video summarization.

Our approach to feature extraction is guided by the observation that behavioural significance in human communication often emerges from the alignment of cues across multiple channels, such as a nod coinciding with vocal emphasis and key terminology, frequently signals important content. We, therefore, design our extraction pipelines to not only capture modality-specific features effectively but also to preserve temporal relationships that enable subsequent cross-modal integration. This dual approach enables us to compare the efficacy of interpretable heuristic methods against more complex neural approaches in behaviour-aware summarization.

3.4.1. Visual Feature Extraction

We implement a multi-faceted visual processing pipeline designed to capture both explicit behavioural signals and semantic visual content that contribute to narrative significance. Our approach focuses particularly on facial and head movements, as these non-verbal cues frequently accompany emphasized speech in conversational settings and provide valuable indicators of communicative intent.

Our visual analysis focuses particularly on facial and head movements, as these non-verbal cues frequently accompany emphasized speech in conversational settings.

We observe that in the interview context of the ChaLearn dataset, speakers often emphasize important points through subtle head movements, facial expressions, and changes in emotional display. By detecting these behavioural markers and integrating them with semantic visual features, we establish a foundation for identifying moments of multimodal significance in the video content.

Head Movement Detection

We develop a landmark-based tracking approach to identify frames exhibiting significant head or facial motion, which often signify communicative emphasis or engagement in interview settings. We employ MediaPipe Pose [13] to track facial landmarks across consecutive frames, capturing subtle but meaningful movements such as nodding, tilting, or turning that frequently accompany important verbal statements. This landmark-based tracking approach enables us to identify frames exhibiting significant head or facial motion, which often signify communicative emphasis or engagement in interview settings.

MediaPipe [13], a lightweight and accurate framework, detects facial landmarks (e.g., nose, eyes) as 3D coordinates, normalized to frame dimensions. Configured for high-confidence detection (minimum 0.5), it processes RGB frames efficiently (<10 ms per frame on CPU), enabling scalability for the dataset’s approximately 30,000 frames. Unlike MediaPipe Face Mesh’s 468 landmarks, Pose’s subset is sufficient for head movement detection, balancing precision and efficiency for interview settings. The detection process identifies frames with notable head or facial motion, capturing non-verbal cues in the ChaLearn dataset’s interview context. For efficient detection, we quantify motion using the mean Euclidean distance between corresponding landmarks across consecutive frames:

$$\text{Movement} = \frac{1}{M} \sum_{i=1}^M \sqrt{(x_{t,i} - x_{t-1,i})^2 + (y_{t,i} - y_{t-1,i})^2 + (z_{t,i} - z_{t-1,i})^2} \quad (1)$$

where M represents the number of tracked landmarks (usually 10 points including nose tip and eye corners), and $(x_{t,i}, y_{t,i}, z_{t,i})$ and $(x_{t-1,i}, y_{t-1,i}, z_{t-1,i})$ are the 3D coordinates of the i -th landmark at frames t and $t-1$ respectively. We implement an adaptive thresholding method to accommodate variations in speaker expressiveness across different videos:

$$\text{Threshold}_t = \mu_t + \phi \cdot \sigma_t \quad (2)$$

where μ_t and σ_t are the mean and standard deviation of movement values over t (a 5-frame window), and ϕ is a scaling factor determined empirically. This adaptive approach allows us to identify significant movements relative to each speaker’s baseline expressiveness, rather than applying a fixed threshold that might miss subtle cues from less expressive speakers or over-sensitize to naturally animated ones.

Both frameworks utilize these head movement cues as illustrated in Figure 6, with Framework I directly incorporating them as signals for *bonus words* identification, while Framework II integrates them with CLIP-derived visual features before transformer-based encoding. By capturing these subtle movements, we establish

a foundation for detecting moments where non-verbal communication reinforces or enhances verbal content, providing valuable signals for summary generation.

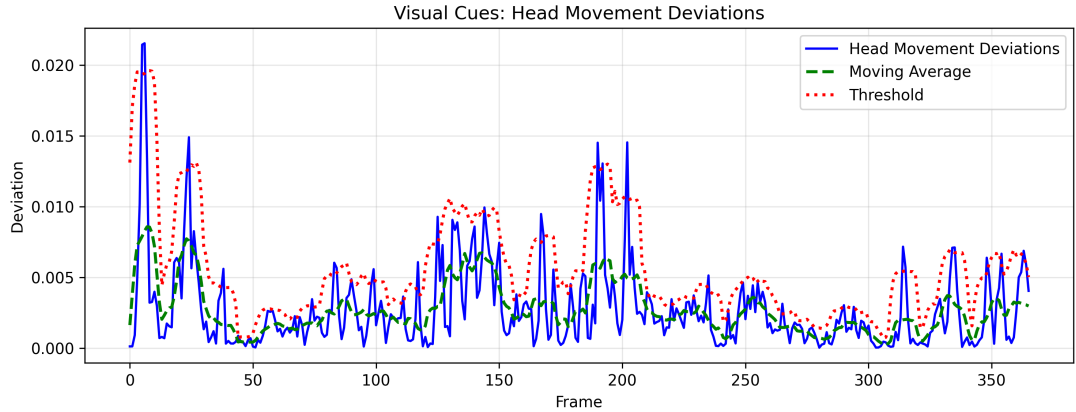


Figure 6. **Example of head movement detection for behavioural visual cues identification.** The graph displays frame-by-frame head position deviations (blue line) measured as Euclidean distance between consecutive frames using facial landmark tracking. The moving average (green dashed line) smooths the signal, while the adaptive threshold (red dotted line) identifies significant movements.

Emotional Shift Detection

We implement emotion detection using the DeepFace framework [14] to identify frames where speakers transition between emotional states, capturing moments of expressive significance that often coincide with important content. This approach focuses on detecting emotional shifts rather than static emotional states, as transitions typically signal moments of narrative importance in interview contexts.

DeepFace [14], a robust facial expression recognition framework, utilizes pretrained convolutional neural networks to detect faces and classify emotions into seven categories: happy, sad, angry, surprised, fear, disgust, and neutral. It processes RGB frames by extracting the facial region and producing a probability distribution across emotion categories, with the highest-probability emotion designated as dominant. Processing is efficient, typically under 100 milliseconds per frame on a standard CPU. Configured for single-face detection to suit the interview setting, DeepFace employs a high-confidence threshold for face localization. Its pretrained models, trained on diverse datasets, ensure robustness across lighting conditions, facial angles, and demographics.

We tailor our implementation to identify expressive transitions that mark important moments in the dataset’s interview context, such as a smile during an enthusiastic response or a frown when addressing challenging topics. The process comprises three key steps: First, we analyze each frame to detect the speaker’s face and determine the dominant emotion based on probability distribution across emotion categories. Second, we monitor emotions over a 5-frame window (5 seconds at 1 fps), with a transition flagged when distinct emotions appear, indicating a shift (e.g., neutral to happy). Finally, we timestamp frames showing emotional transitions and mark them as significant, facilitating integration with other modalities for summarization.

We design our temporal analysis to track emotions within a 5-frame window, selected to capture meaningful transitions while filtering out noise. A significant transition is identified when at least two distinct emotions appear within the window, such as the sequence [neutral, neutral, *happy*, happy, happy], where the third frame would be flagged as the transition point. This approach emphasizes expressive shifts, such as a smile during a key statement, which are important for identifying emotionally significant moments. To maintain reliability, we exclude frames where face detection fails from analysis, preventing misclassification that could arise from partial or occluded faces.

Both frameworks incorporate these emotional transition cues, with Framework I using them directly as indicators for *bonus words* detection, while Framework II integrates them with other visual features as part of its richer representation. This approach allows us to detect moments where emotional expression reinforces verbal content, providing another dimension for identifying key moments across modalities for summarization.

Semantic Visual Features

We implement semantic visual feature extraction using the Contrastive Language-Image Pre-training (CLIP) model [15] for Framework II specifically, augmenting explicit behavioural cues with deep semantic representations. This approach provides a richer understanding of visual content beyond behavioural signals, capturing semantic elements that contribute to narrative significance.

CLIP [15], developed by OpenAI, is a multimodal model trained on 400 million image-text pairs, creating a shared embedding space where images and text are semantically aligned. It consists of an image encoder which is a vision transformer (ViT) model that processes images to produce 768-dimensional embeddings, capturing visual elements such as objects, textures, and contextual information. CLIP leverages contrastive representation learning, where similar samples, such as an image and its matching caption, are mapped close together in embedding space, while dissimilar samples are pushed apart. Unlike standard contrastive learning [144], which uses triplets of (anchor, positive, negative), CLIP employs a multi-class N -pair loss that generalizes by considering multiple positives and negatives for each anchor within a batch, enhancing efficiency and robustness. In the multi-class N -pair loss, for a batch of (N) image-text pairs, each image serves as an anchor, its corresponding caption as the positive, and the $(N - 1)$ other captions as negatives (and vice versa for text). The loss function maximizes the cosine similarity between matched image-text pairs while minimizing it for unmatched pairs, formalized as:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \left[\log \frac{\exp(\text{sim}(I_i, T_i)/\omega)}{\sum_{j=1}^N \exp(\text{sim}(I_i, T_j)/\omega)} + \log \frac{\exp(\text{sim}(T_i, I_i)/\omega)}{\sum_{j=1}^N \exp(\text{sim}(T_i, I_j)/\omega)} \right] \quad (3)$$

where I_i and T_i are the image and text embeddings for the (i) -th pair, $\text{sim}(\cdot, \cdot)$ is cosine similarity, and ω is a temperature parameter. This approach enables CLIP to efficiently produce robust embeddings across diverse visual and textual inputs, making it ideal for extracting semantic features from video frames in the summarization framework.

CLIP is selected over other vision-language models (VLMs) such as ViLBERT [145], BLIP [146], or Flamingo [147] for its efficiency and alignment with our proposed approach, better generalization and easy use of OpenAI’s pre-trained weights via HuggingFace integration. CLIP’s 400 million image-text pairs provide superior generalization compared to ViLBERT’s smaller datasets, ensuring robust embeddings across the ChaLearn dataset’s varied settings. The multi-class N-pair loss enables zero-shot tasks, allowing dynamic criteria (e.g., frames with expressive cues) without fine-tuning, unlike ViLBERT’s task-specific requirements. Also, CLIP’s vision transformer (ViT) processes frames in almost 50 ms on a GPU, more efficient than BLIP’s generative models. CLIP’s contrastive alignment facilitates integration with audio and text, unlike Flamingo’s vision-centric focus.

For video summarization, CLIP’s image encoder processes frames to produce 768-dimensional embeddings that capture semantic content (e.g., speaker’s appearance, interview setting), ensuring embeddings reflect both visual and behavioural significance. The novel approach integrates head movement and emotion cues to enhance CLIP embeddings, ensuring they reflect both semantic visual content and behavioural dynamics for improved summarization outcomes. This integration improves embedding quality by focusing CLIP’s semantic power on frames with dynamic behavioural cues, such as a head nod during a confident statement or a smile signalling enthusiasm, resulting in embeddings that better represent key interview moments.

We integrate CLIP into our framework through a streamlined process that extracts features from video frames and combines them with behavioural cues. First, we sample frames at 1 fps to balance computational efficiency with temporal granularity. Next, we resize each frame to 224x224 pixels and normalize RGB values to [0, 1] scale to match CLIP’s ViT input requirements. We then process each frame through CLIP’s image encoder (ViT-B/32), producing a 512-dimensional embedding that captures semantic visual content. These embeddings encode information about the visual context, including the speaker’s appearance, posture, and environmental elements. We enhance these embeddings by concatenating them with binary indicators from our facial movement and emotional shift detection, creating a comprehensive 514-dimensional visual representation that combines semantic understanding with explicit behavioural cues.

Only Framework II utilizes these deep semantic embeddings, while Framework I maintains its focus on directly interpretable behavioural cues for *bonus words* identification. This distinction represents a key difference in feature richness between the two approaches, allowing us to evaluate the impact of deeper semantic understanding versus more transparent behavioural signals for video summarization.

3.4.2. Audio Feature Extraction

We implement a comprehensive audio processing pipeline to capture the expressive elements of speech that convey behavioural intent and emotional states. Our approach combines traditional paralinguistic feature extraction with advanced contextual representations, providing a multi-level analysis of speech dynamics that complements our visual processing pipeline.

We observe that in interview contexts, speakers often mark important content through vocal modulation, such as raising pitch during enthusiastic statements or increasing volume for emphasis. By detecting these acoustic patterns and integrating them with semantic audio features, we establish another dimension for identifying behaviourally significant moments across modalities.

Prosodic Features

We implement prosodic feature extraction using the Yet Another Algorithm for Pitch Tracking (YAAPT) [16] to capture fundamental frequency (F_0) variations that reflect intonation and expressiveness in speech. This approach enables us to identify patterns such as rising pitch during questions or emphasized statements, which frequently signal important content in interview contexts.

YAAPT [16] leverages a hybrid spectral-temporal approach to estimate the fundamental frequency, crucial for capturing intonation and expressiveness in speech. The algorithm combines normalized cross-correlation in the frequency domain to identify pitch candidates by analyzing harmonic patterns, with dynamic programming to optimize the pitch contour by minimizing discontinuities across frames. This approach ensures robust pitch tracking in the presence of background noise and maintains accurate tracking through unvoiced segments where pitch estimation is inherently challenging. The objective of YAAPT can be formalized as finding the optimal pitch trajectory $F_0(t)$ that maximizes the likelihood of spectral harmonics while enforcing temporal continuity:

$$F_0(t) = \arg \max_f \sum_t [\text{NCC}(s(t), f) + \theta \cdot \text{smoothness}(f(t), f(t-1))] \quad (4)$$

where NCC is the normalized cross-correlation score, and smoothness penalizes abrupt pitch changes, weighted by a regularization parameter, θ . We select YAAPT over alternatives such as Praat [148] or CREPE [149] for its noise robustness and dynamic programming capabilities. Praat’s heuristic-based approach proves less consistent in interview settings with potential background noise, while CREPE’s neural model introduces computational overhead without proportional gains for our short interview clips. YAAPT’s hybrid approach provides the optimal balance of accuracy and efficiency for our dataset.

We configure YAAPT with parameters optimized for conversational speech analysis: a frame length of 35ms to capture complete pitch periods even for low-pitched voices, a frame step of 10ms to provide sufficient temporal resolution for detecting prosodic changes, and a pitch range of 60-400Hz to accommodate both male and female speakers in our dataset. We enable the nonlinear dynamic programming algorithm with a continuity weight of 0.5 to balance pitch accuracy with temporal smoothness.

From YAAPT’s output, we extract the interpolated pitch contour ($f0_samp_interp$), replacing NaN values in unvoiced segments with the median pitch to ensure continuity. We compute both frame-level values for detailed analysis and aggregated statistics, including the mean pitch as a scalar feature:

$$\text{mean } F_0 = \frac{1}{N} \sum_{t=1}^N f0_samp_interp(t) \quad (5)$$

where N is the number of frames in the audio segment. We also calculate pitch variation metrics such as standard deviation and range to capture dynamic intonation patterns. Pitch captures prosodic cues, such as rising intonation during enthusiastic responses, important for identifying narratively significant moments. Its robustness to noise ensures reliability in varied interview conditions, unlike simpler pitch trackers (e.g., autocorrelation), which falter in unvoiced segments.

Spectral Features

We extract spectral features using the openSMILE toolkit [17] with the extended Geneva Minimalistic Acoustic Parameter Set (eGeMAPS) to capture voice quality and perceptual intensity variations that often signal emotional states or emphasized speech. This approach complements our prosodic analysis by providing insights into the spectral characteristics of speech that convey paralinguistic information.

The openSMILE toolkit [17] with eGeMAPS offers a comprehensive set of standardized acoustic features specifically designed for paralinguistic analysis. From this extensive feature set, we select two parameters, such as the Hammarberg Index to quantify voice quality or tonality, and Loudness to represent perceptual intensity. These parameters were chosen after correlation analysis showed them to be most strongly associated with perceived emphasis in our interview context.

Hammarberg Index. The Hammarberg index is a spectral feature that quantifies the balance of energy between low and high frequency bands in the speech signal, serving as a robust indicator of voice quality, such as breathiness, tenseness, or vocal effort. It is mathematically defined as:

$$\text{Hammarberg} = 10 \cdot \log_{10} \left(\frac{\int_0^2 E(f), df}{\int_2^5 E(f), df} \right) \quad (6)$$

where $E(f)$ represents the spectral energy density at frequency (f) , $(\int_0^2 E(f), df)$ is the energy in the 0-2 kHz band (low frequencies, encompassing fundamental and lower formants), and $(\int_2^5 E(f), df)$ is the energy in the 2-5 kHz band (higher frequencies, associated with articulation and noise components). The ratio, expressed in decibels (dB) via the logarithmic scale, highlights the relative dominance of low-frequency energy, with higher values indicating a breathier or softer voice (more low-frequency energy) and lower values suggesting a tenser or sharper voice (more high-frequency energy). This makes the Hammarberg Index particularly sensitive to paralinguistic cues, such as emotional states or speaker affect, which are important for identifying narrative significance in interview audio.

The significance of the Hammarberg Index lies in its ability to capture essential aspects of voice quality that correlate with perceptual characteristics. A higher Hammarberg Index typically indicates increased energy in higher frequencies relative to lower frequencies, which can be associated with a brighter, sharper, or more strident voice quality. Conversely, a lower index suggests relatively more energy in the lower frequency bands, often perceived as a darker, warmer, or more muffled voice quality.

This parameter is particularly effective for detecting breathiness and variations in vocal effort that signal affective states in interview contexts. Breathier voices tend to

show specific patterns in the Hammarberg Index due to their characteristic spectral tilt and energy distribution. Similarly, changes in vocal effort (from soft to loud phonation) produce systematic changes in the spectral balance that are reflected in the Hammarberg Index values. In clinical and research settings, this measure has proven valuable for differentiating between normal and pathological voices, as many voice disorders manifest as alterations in spectral energy distribution.

OpenSMILE’s eGeMAPSv02 feature set extracts frame-level Hammarberg index values (*hammarbergIndex_sma3*), computed with a smoothed moving average (SMA) over three frames to reduce noise. These values are averaged across all frames to yield a single scalar per clip:

$$\text{Hammarberg} = \frac{1}{N} \sum_{n=1}^N \text{hammarbergIndex_sma3}(n) \quad (7)$$

where N is the number of frames. The Hammarberg index enhances the audio representation by capturing voice quality variations that signal affective states, such as breathiness during emotional or hesitant speech or tenseness during assertive responses, which are pivotal for narrative-driven summarization. For example, a high Hammarberg index may indicate a softer, emotionally charged moment in an interview, aligning with visual cues like expressive gestures. Its spectral focus complements the prosodic (pitch) and perceptual (loudness) features, providing a multidimensional view of the speaker’s vocal characteristics.

Loudness. It represents the perceptual intensity of sound as experienced by human listeners, making it fundamentally different from physical sound pressure level or amplitude measurements. The computation of loudness in openSMILE implements sophisticated psychoacoustic modeling that accounts for the complex, non-linear nature of human auditory perception. We also extract Loudness based on Zwicker’s psychoacoustic model [150], which accounts for the frequency-dependent sensitivity of human hearing across critical bands:

$$\text{Loudness} = \sum_b w_b \cdot E_b \quad (8)$$

where w_b are band-specific weights, and E_b is the energy in critical bands. This perceptual measure incorporates the non-linear relationship between physical intensity and perceived loudness, making it more representative of human auditory perception than simple amplitude or energy measurements.

The loudness measure incorporates several key aspects of human hearing: it accounts for the frequency-dependent sensitivity of the human ear, which is more sensitive to mid-range frequencies (roughly 2-5 kHz) and less sensitive to very low and very high frequencies, and it considers the non-linear relationship between physical intensity and perceived loudness, where a doubling of physical sound intensity does not result in a doubling of perceived loudness.

In spectral analysis, loudness serves as a crucial parameter for characterizing vocal expression, emotion, and emphasis. Various emotional states manifest through distinct loudness patterns, such as anger often features high loudness, while sadness typically

shows reduced loudness. Within speech, variations in loudness highlight emphasis and structure, marking important words or phrases. These loudness patterns extend beyond simple amplitude changes, capturing nuanced aspects of vocal expression that simple energy measurements would miss. When combined with temporal analysis, loudness provides insights into speech rhythm, stress patterns, and dynamic range, all of which are essential components of natural and expressive communication. For instance, loudness peaks highlight stressed syllables or emphatic speech, while dynamic range (variation between soft and loud segments) reflects emotional shifts, such as intensity shifts during passionate responses. The mean loudness value calculated across an utterance offers a global characterization of the overall perceived intensity, while other statistical measures (range, standard deviation) can capture the dynamic variation in vocal intensity that contributes to expressiveness and emotional colouring of speech.

OpenSMILE’s eGeMAPSv02 extracts frame-level loudness values (*Loudness_sma3*) with a three-frame smoothed moving average to mitigate noise, averaged across frames to produce a scalar per clip:

$$\text{Loudness} = \frac{1}{N} \sum_{n=1}^N \text{Loudness_sma3}(n) \quad (9)$$

where N is the number of frames. Loudness enhances the audio representation by capturing perceived intensity, important for detecting narrative emphasis in interviews, such as louder speech during confident or emotional moments. Combined with temporal analysis, loudness reveals speech rhythm (e.g., alternating loud and soft segments in conversational flow), stress patterns (e.g., accented words in key statements), and dynamic range (e.g., transitions from quiet to loud for dramatic effect), all contributing to the expressiveness and emotional colouring of speech. For example, a high mean loudness may indicate an assertive interview response, complementing visual cues like head nods. However, the mean-based aggregation limits insights into dynamic variation, which other statistical measures (range, standard deviation) could capture to further highlight expressiveness.

Both frameworks incorporate these prosodic and spectral features as illustrated in Figure 7, with Framework I using significant variations as direct indicators for *bonus words* identification, while Framework II integrates them with other audio features in its richer representation. These features provide additional dimensions for capturing vocal expressiveness, complementing the visual and textual features to form a comprehensive representation for summarization.

Contextual Audio Representation

We implement contextual audio representation using the Hidden-Unit BERT (HuBERT) model [18] for Framework II specifically, enhancing traditional acoustic features with deep contextual embeddings. This approach captures rich temporal dynamics in speech through a masked prediction paradigm, providing embeddings that reflect both linguistic content and paralinguistic cues.

The HuBERT framework represents a significant advancement in self-supervised speech representation learning, providing contextual embeddings that capture the multidimensional nature of speech signals. Unlike traditional acoustic feature

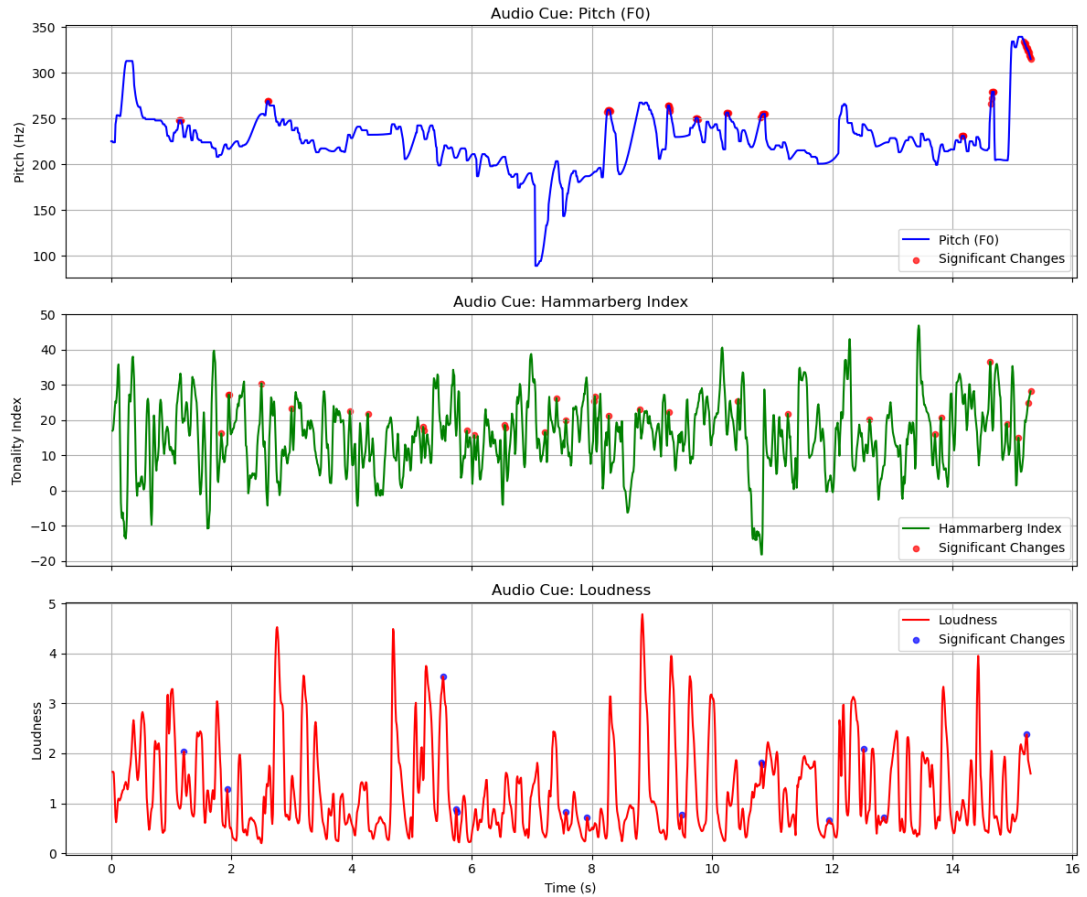


Figure 7. **Examples of prosodic and spectral features for audio cues detection.** The figure shows three acoustic features extracted from an audio: (top) fundamental frequency/pitch (F_0) tracking intonation patterns; (middle) Hammarberg Index measuring voice quality (spectral slope); and (bottom) loudness measurements capturing speech intensity, with significant audio cues marked as round.

extraction methods that focus on predetermined paralinguistic features, HuBERT leverages a masked prediction approach similar to BERT [151] in natural language processing. It begins by clustering acoustic features into discrete speech units, which serve as pseudo-labels for a prediction task. The model masks segments of the audio and learns to infer these hidden units from surrounding context, building an understanding of speech as a dynamic interplay of phonetic, prosodic, and speaker-specific elements.

The architecture of HuBERT implements a multi-stage training process that begins with an initial clustering of acoustic features to establish pseudo-labels. These discrete units serve as targets for a masked prediction task, wherein the model learns to predict hidden segments of speech from surrounding context. This process enables the model to develop a nuanced understanding of the temporal dependencies and contextual relationships within speech signals. As training progresses through multiple iterations, the quality of these pseudo-labels improves, resulting in increasingly refined representations. A distinctive advantage of HuBERT lies in its ability to differentiate between content-dependent and content-independent aspects of speech.

By learning from vast quantities of unlabeled speech data spanning diverse speakers and acoustic conditions, the model develops robust internal representations that effectively disentangle speaker identity, emotional state, and linguistic content. This capability proves invaluable for paralinguistic analysis, where the objective often involves isolating specific voice characteristics from the underlying phonetic content.

We select HuBERT over alternatives, such as Wav2Vec 2.0 [152] or Whisper [142] for several compelling reasons. HuBERT’s self-supervised learning approach is particularly well-suited for conversational audio, providing rich representations without requiring labeled data. In contrast, Wav2Vec 2.0’s supervised pretraining on labeled data introduces limitations for diverse speech patterns, potentially requiring additional fine-tuning for our interview context. Whisper, while excellent for transcription, prioritizes textual output over rich embeddings and introduces computational overhead. HuBERT’s masked prediction of clustered speech units fosters a nuanced understanding of speech dynamics, balancing expressiveness and efficiency to complement our visual feature extraction pipeline.

We integrate HuBERT into our framework through a multi-stage processing pipeline. First, we normalize the raw audio waveform to a consistent amplitude range, ensuring robustness across varied recording conditions. We then resample the audio to 16 kHz to match HuBERT’s training configuration and process it through the HuBERT-base-ls960 model to generate 768-dimensional frame-level embeddings:

$$h_t = \text{HuBERT}(x_t; \Theta) \in \mathbb{R}^{768}, \quad t = 1, \dots, N \quad (10)$$

where x_t is a waveform segment, Θ is a pre-trained parameter, and N is the number of frames. The transformer’s positional encodings ensure each embedding captures not just local acoustics but also contextual relationships, such as a phoneme’s role in a stressed syllable versus a casual phrase. This hierarchical structure, such as lower layers encoding phonetic details, higher layers abstracting prosodic and contextual patterns, enables HuBERT to disentangle linguistic content (e.g., words spoken) from paralinguistic cues (e.g., emotional tone), making it a powerful tool for summarization. These embeddings capture a hierarchical representation of speech, with lower layers encoding phonetic details and higher layers abstracting prosodic and contextual patterns.

Only Framework II utilizes these deep contextual representations, while Framework I maintains its focus on directly interpretable acoustic features for *bonus words* identification. This distinction represents another key difference in feature richness between the two approaches, allowing us to evaluate the impact of deeper contextual understanding versus more transparent acoustic features for video summarization.

3.4.3. Textual Feature Extraction

We implement a multi-faceted text processing pipeline to identify both semantic significance and structural importance within the transcript. This component complements our visual and audio analysis by capturing the linguistic patterns and discourse structure that contribute to narrative significance in interview contexts.

We observe that in interview scenarios, important content is often marked by distinctive terminology, emotionally charged language, or references to specific entities. By detecting these textual patterns and integrating them with behavioural cues from other modalities, we establish another dimension for identifying significant moments for summarization.

Semantic Textual Analysis

We implement Term Frequency-Inverse Document Frequency (TF-IDF) analysis [30] to identify content-rich terms that distinguish specific segments from general discourse. This statistical approach enables us to highlight words and phrases that carry distinctive information relative to the broader corpus, providing valuable signals for detecting semantically significant content. TF-IDF [30] quantifies the importance of terms within a document relative to a corpus by combining term frequency (how often a term appears in a document) with inverse document frequency (how rare the term is across all documents). This approach assigns higher weights to terms that appear frequently in a specific document but rarely in the overall corpus, highlighting distinctive terminology that often represents key concepts or focus points. For each term t in document d within corpus D , we compute:

$$\text{TF-IDF}(t, d, D) = \text{TF}(t, d) \cdot \text{IDF}(t, D) \quad (11)$$

where $\text{TF}(t, d)$ is the term frequency of t in d , normalized by document length to prevent bias toward longer documents, and $\text{IDF}(t, D) = \log \frac{|D|}{|\{d \in D: t \in d\}|}$ is the inverse document frequency, reflecting the term’s rarity across the corpus.

Our TF-IDF implementation provides finer control over preprocessing steps, enables incorporation of domain-specific stopword lists tailored to interview contexts, and it allows for specialized weighting schemes that better capture semantic significance in short texts. Our implementation adapts the traditional TF-IDF formula to better handle the short document length of interview responses by incorporating a length normalization factor that prevents bias toward longer sentences. Our preprocessing pipeline includes multiple stages, such as tokenization using SpaCy [153] for accurate boundary detection, removal of custom stopwords including fillers common in conversational speech (e.g., "um", "like", "you know"), lemmatization to normalize morphological variants, and case normalization. These preprocessing steps are specifically optimized for conversational transcripts, where standard approaches might miss important nuances.

We implement TF-IDF at both the sentence and document levels, enabling identification of terms that distinguish specific sentences within a response (local significance) and terms that distinguish the entire response within the corpus (global significance). Terms with high TF-IDF scores (usually in the top 20th percentile) are considered semantically significant and given higher weight in our summarization process, with scores normalized to a 0-1 range for consistent integration with other features.

In Framework I, terms with high TF-IDF scores are prioritized as *bonus words* candidates, directly influencing sentence selection to ensure summaries capture

distinctive content. Framework II does not utilize TF-IDF, relying instead on its transformer-based architecture for feature integration.

Sentiment Analysis

We perform sentiment analysis to identify emotionally charged language that often signals important content or narrative transitions. This approach enables us to detect segments where speakers express strong opinions or emotional responses, which frequently correspond to content they consider particularly significant.

We implement a hybrid sentiment analysis approach that combines lexicon-based scoring with contextual adjustments. Our sentiment analyzer uses the NRC Emotion Lexicon [154] as its foundation, which provides fine-grained emotional categorization beyond simple positive/negative polarity. This lexicon maps words to eight basic emotions (anger, fear, anticipation, trust, surprise, sadness, joy, and disgust) as well as positive and negative sentiment, enabling nuanced emotional analysis of interview transcripts.

We select the NRC Emotion Lexicon [154] over alternatives such as VADER [155] and TextBlob [156] for its robust performance in sentiment and emotion analysis across diverse textual data. Unlike VADER, which is optimized for social media text, and TextBlob, which provides general-purpose sentiment analysis, the NRC lexicon offers a comprehensive mapping of words to emotional categories, making it well-suited for capturing nuanced affective cues in the conversational transcripts of our behaviour-aware multimodal video summarization framework. We extend the basic lexicon matching with contextual rules for handling negations (e.g., "not happy"), intensifiers (e.g., "very sad"), and attenuators (e.g., "somewhat surprised"). Specifically, we implement a five-word lookback window to detect negation words that invert sentiment polarity, and we apply multiplicative scaling for intensifiers (e.g., "very" multiplies sentiment by 1.5) and attenuators (e.g., "slightly" multiplies sentiment by 0.5). These contextual adjustments significantly improve accuracy for conversational speech, where sentiment is often expressed with such modifiers.

We calculate sentiment at both the word and sentence levels. At the word level, each term is assigned a sentiment score based on the lexicon, modified by contextual rules. At the sentence level, we compute an aggregate sentiment score as the sum of word-level scores, normalized by sentence length to prevent bias toward longer sentences. Our implementation produces normalized scores between -1 (strongly negative) and $+1$ (strongly positive), with additional categorical labels for the eight basic emotions in the NRC lexicon. We identify sentences with high positive or negative sentiment intensity (absolute value ≥ 0.6) as emotionally significant, as these often contain emphatic statements or emotionally charged content. Words in these high-intensity sentences receive additional weight in our summarization process, reflecting their potential narrative importance. Through empirical analysis, we observe that emotionally charged language frequently coincides with content that speakers consider important or noteworthy.

In Framework I, high-scoring terms from these sentences are used as *bonus words* candidates to guide summary generation. Framework II does not incorporate sentiment analysis, relying solely on its transformer-based architecture for textual feature processing.

Named Entity Recognition

We employ Named Entity Recognition (NER) [137] to identify key entities such as people, organizations, and locations, which serve as important anchor points in the narrative structure of interview transcripts. These entities carry high informational value, ensuring coherence and comprehensibility in generated summaries by preserving essential contextual references.

We use SpaCy’s small English model (*en_core_web_sm*) [143] for NER, selected for its balance of efficiency and robust performance in detecting entity types such as PERSON, ORG, and GPE (geo-political entity) in conversational data. SpaCy was selected over alternatives such as NLTK’s NER module and Stanford NER [157] for its robust performance and adaptability to conversational data, making it well-suited for our multimodal video summarization framework. To enhance entity detection, we implement a custom post-processing step that resolves coreferences (e.g., linking pronouns to their antecedent entities) and normalizes variant references (e.g., treating "Google" and "Google Inc." as the same entity), improving consistency across transcripts.

Named entities act as contextual anchors, often signaling segments where speakers introduce significant concepts or examples. By prioritizing these entities, we ensure summaries retain important information for narrative clarity. In Framework I, these entities are prioritized as *bonus words* candidates, directly influencing sentence selection. Framework II does not utilize NER, relying instead on its transformer-based architecture for feature integration.

Contextual Text Embeddings

We implement contextual text embeddings using the RoBERTa model [19] for Framework II specifically, enhancing traditional textual analysis with deep contextual representations. This approach, illustrated in Figure 8, captures subtle semantic relationships and discourse structure that might be missed by more explicit feature extraction methods.

RoBERTa [19] (Robustly Optimized BERT Pre-training Approach) represents a significant advancement over the original BERT model, implementing improved training methodologies including dynamic masking, larger batch sizes, and longer training sequences. These enhancements enable RoBERTa to achieve state-of-the-art performance on various NLP benchmarks, learning contextual representations that effectively capture word meaning in context and handle linguistic ambiguities.

We select RoBERTa [19] over alternatives such as BERT [151] and DistilBERT [158] for its enhanced performance in natural language processing tasks relevant to semantic understanding. RoBERTa’s optimized pretraining, leveraging larger datasets and refined training objectives, consistently yields superior contextual embeddings compared to BERT’s foundational transformer architecture and DistilBERT’s computationally efficient distilled model. This makes RoBERTa particularly suitable for capturing nuanced semantic and behavioural cues in our multimodal video summarization framework.

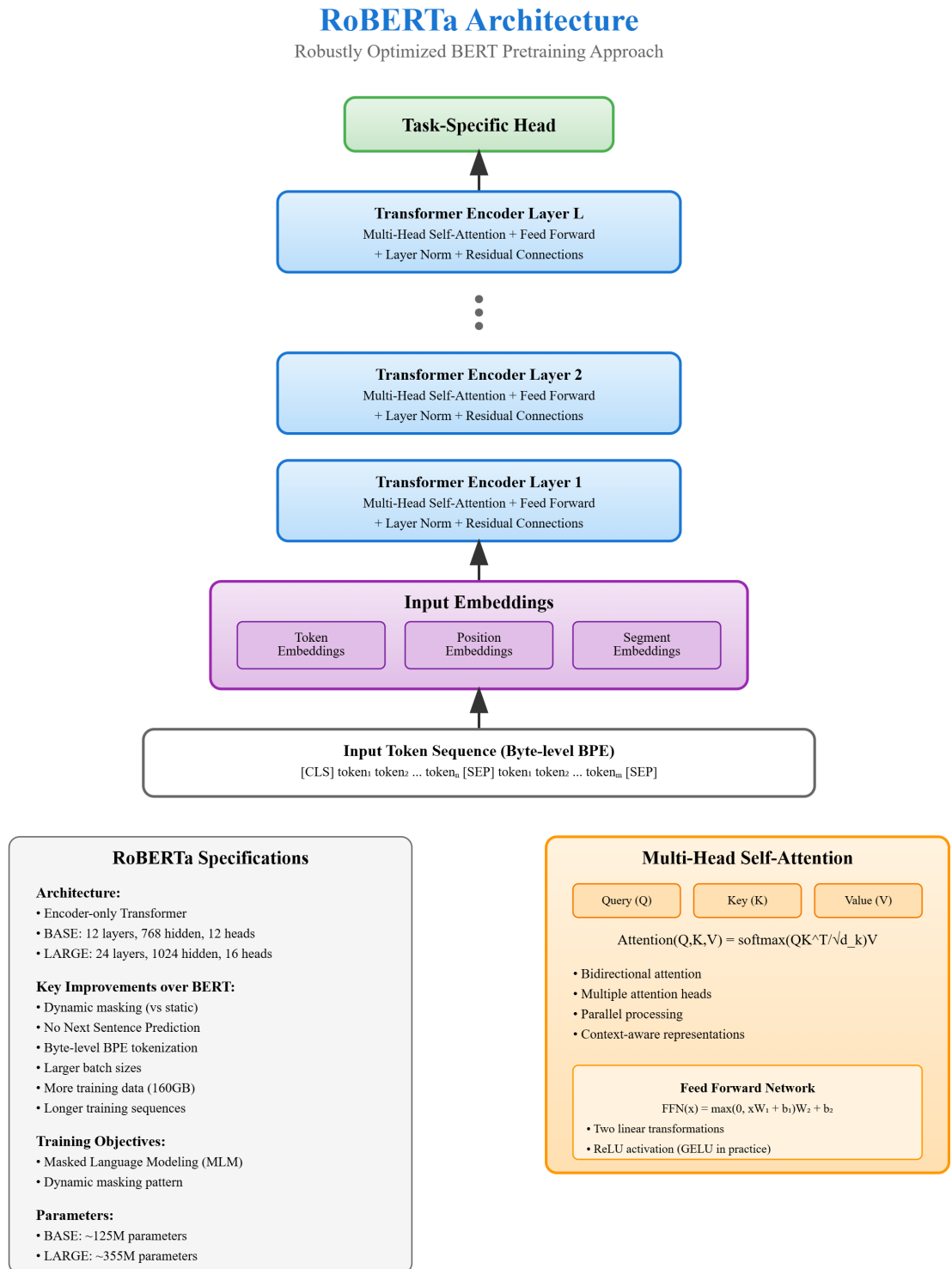


Figure 8. **The graphical representation of RoBERTa (Robustly Optimized BERT Pretraining Approach) architecture, inspired from [19].** This diagram illustrates the encoder-only transformer architecture of RoBERTa, showing the flow from input tokenization through multiple transformer encoder layers to task-specific outputs.

We utilize RoBERTa through a SentenceTransformer⁵ implementation fine-tuned specifically for semantic similarity tasks. This model generates 1024-dimensional

⁵[sentence-transformers/all-roberta-large-v1](https://github.com/UKPLab/sentence-transformers/tree/master/all-roberta-large-v1)

embeddings that effectively capture sentence-level semantics and their relationships, providing a rich representation of the transcript’s linguistic content. The model has been trained on diverse text corpora using contrastive learning objectives, enabling it to map semantically similar sentences closer together in the embedding space. We implement a hierarchical encoding process that begins by segmenting the transcription text into individual sentences, usually 3–5 per clip due to the brevity of the interviews. Each sentence is processed by the SentenceTransformer to generate 1024-dimensional embeddings:

$$s_i = \text{SentenceTransformer}(T_i) \in \mathbb{R}^{1024}, \quad i = 1, \dots, k \quad (12)$$

where T_i is the (i)-th sentence in the transcript, and k is the number of sentences. These embeddings capture not only the internal semantic structure of each sentence but also its potential contextual relationships with other sentences.

Only Framework II utilizes these deep contextual embeddings, while Framework I maintains its focus on directly interpretable textual features for *bonus words* identification. This distinction represents another key difference in feature richness between the two approaches, allowing us to evaluate the impact of deeper semantic understanding versus more transparent textual features for video summarization.

3.5. Summarization Frameworks

We implement two complementary summarization frameworks that process the extracted multimodal features to generate behaviour-aware video summaries. These frameworks represent fundamentally different approaches to the same challenge: Framework I employs a transparent, and rule-based methodology that identifies *bonus words* emphasized across multiple modalities, while Framework II leverages transformer-based architectures with cross-modal attention mechanisms to generate summaries through autoregressive decoding.

Our dual-framework approach enables comprehensive evaluation of different computational paradigms for behaviour-aware video summarization. Framework I prioritizes interpretability and computational efficiency, making its decision-making process directly observable and understandable [27]. Framework II emphasizes representational power and semantic depth, leveraging state-of-the-art transformer-based architecture to capture complex patterns across modalities [11]. By comparing these approaches, we gain insights into the trade-offs between interpretability and performance, as well as the relative importance of explicit behavioural cues versus learned semantic representations.

3.5.1. Framework I: Heuristic Approach

We build Framework I upon the foundational principles of extractive summarization established by Edmundson [27], while significantly extending the approach to incorporate multimodal behavioural analysis. The traditional Edmundson method identifies important sentences based on statistical features such as term frequency,

sentence position, and cue phrases. We enhance this approach by introducing cross-modal emphasis detection as an additional dimension for importance scoring.

Our methodological innovation lies in the systematic integration of behavioural cues across modalities to identify what we term *bonus words*, lexical items that receive coordinated emphasis through visual gestures, audio prosody, and textual significance markers. This integration addresses a fundamental limitation of traditional text-based extractive methods: their inability to capture the rich paralinguistic and non-verbal context that often determines the actual importance of spoken content

We implement our framework through four interconnected processing stages that transform raw multimodal features into coherent video summaries. Each stage incorporates adaptive mechanisms to accommodate individual differences in communication style and expressiveness, ensuring robust performance across diverse speakers and content types.

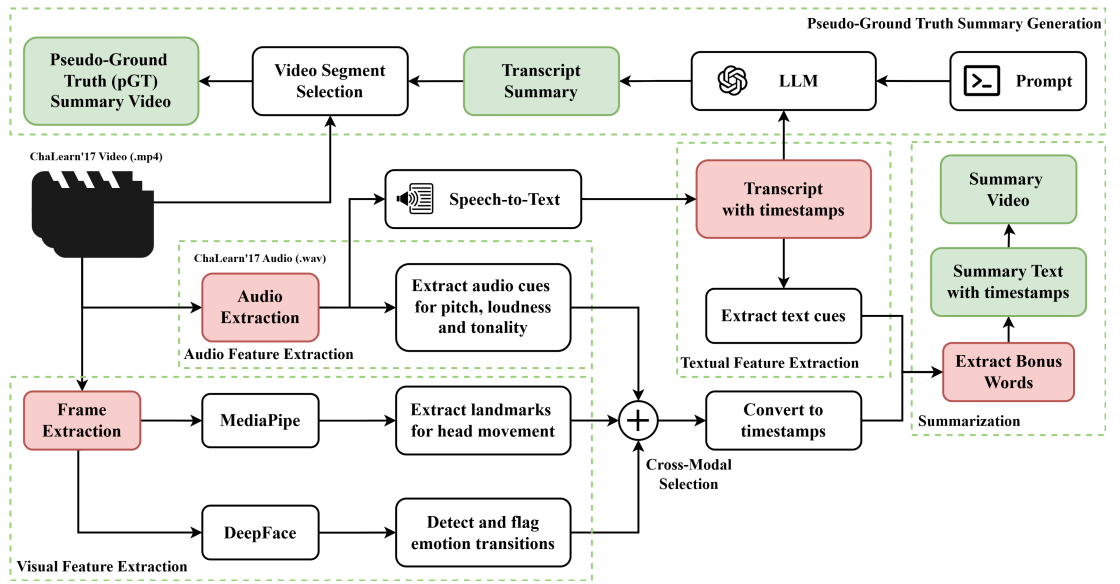


Figure 9. **Framework I: Overall architecture of *bonus words*-based heuristic approach.** This diagram illustrates the complete pipeline for our interpretable multimodal video summarization framework. The process begins with the dataset, followed by parallel feature extraction across all modalities. These features undergo cross-modal selection to identify *bonus words*, which are then used for summarization. The top section shows the pseudo-ground truth generation process using LLM-based extractive summarization for evaluation purposes.

As illustrated in Figure 9, our architecture operates on the principle of detecting moments where multiple modalities exhibit synchronized patterns of emphasis or expressiveness. We begin with temporally aligned outputs from our multimodal feature extraction pipeline, including facial movement indicators, emotional shift markers, prosodic variations, and textual significance scores, and then systematically process these through our four-stage pipeline to generate behaviour-aware summaries.

Cross-Modal Cues Alignment

We implement a sophisticated temporal alignment system that establishes precise correspondence between behavioural cues detected across visual, audio, and textual modalities. This alignment process serves as the foundation for identifying genuine cross-modal emphasis patterns rather than coincidental temporal overlaps.

We establish temporal correspondence through a sliding window approach that examines each word’s timestamp against detected cues from other modalities:

$$\text{Aligned}(w, c_m) = |t_w - t_{c_m}| \leq \tau_{\text{window}} \quad (13)$$

where w represents a word with timestamp t_w , c_m represents a cue from modality m occurring at time t_{c_m} , and $\tau_{\text{window}} = 0.5$ seconds represents our empirically determined alignment window.

For visual cues, we align head movement indicators and emotional transition markers with word timestamps. When multiple visual cues occur within the alignment window, we select the cue with the highest confidence score based on movement magnitude or emotional transition intensity. For audio cues, we align prosodic emphasis markers (pitch variations, loudness changes, voice quality shifts) with corresponding words, applying similar confidence-based selection when multiple cues coincide. We implement confidence weighting for aligned cues to distinguish between strong and weak multimodal correlations:

$$\text{Confidence}(w, c_m) = \exp\left(-\frac{|t_w - t_{c_m}|^2}{2t_r^2}\right) \cdot \text{Magnitude}(c_m) \quad (14)$$

where t_r is the temporal decay rate, and $\text{Magnitude}(c_m)$ represents the strength of the detected cue. This weighting ensures that temporally precise alignments receive higher confidence scores than those occurring at the edge of our alignment window.

Identification of *Bonus Words*

We develop a systematic algorithm for identifying *bonus words* based on cross-modal emphasis patterns, requiring evidence of coordinated emphasis from at least two modalities to ensure reliability while maintaining sensitivity to subtle multimodal patterns.

Our bonus words detection operates through pattern matching across aligned multimodal cues. For each word in the transcript, we examine whether it demonstrates emphasis patterns across multiple modalities within our temporal alignment window. We categorize emphasis evidence into three types: visual (head movements, emotional transitions), auditory (prosodic variations), and textual (statistical significance, sentiment intensity, named entity status). The identification algorithm systematically evaluates each word for multimodal emphasis evidence:

$$\text{Evidence}_{\text{total}}(w) = \sum_{m \in \text{visual, audio, text}} \mathbb{I}(\text{Emphasis}_m(w)) \cdot \text{Confidence}_m(w) \quad (15)$$

where $\mathbb{I}(\text{Emphasis}_m(w))$ is an indicator function for emphasis detection in modality m , and $\text{Confidence}_m(w)$ represents the strength of the detected emphasis pattern.

We establish a multi-modal threshold criterion requiring evidence from at least two modalities:

$$\text{BonusWord}(w) = \begin{cases} \text{True} & \text{if } \sum_m \mathbb{I}(\text{Emphasis}_m(w)) \geq 2 \\ \text{False} & \text{otherwise} \end{cases} \quad (16)$$

Additionally, we implement evidence strength ranking to distinguish between strongly and weakly emphasized bonus words:

$$\text{Rank}(w) = \text{Evidence}_{\text{total}}(w) \cdot \log(1 + \text{NumModalities}(w)) \quad (17)$$

where $\text{NumModalities}(w)$ counts the number of modalities showing emphasis for word, w . This ranking enables our sentence weighting mechanism to differentiate between words with marginal multimodal support and those with strong cross-modal emphasis. We apply additional filtering to remove common function words that might coincidentally align with behavioural cues but lack semantic significance. Our filtered stopword list includes articles, prepositions, and conversational fillers that frequently appear in interview speech but rarely carry emphasis meaning.

Sentence Weighting and Selection for Summarization

We implement a comprehensive sentence importance scoring mechanism, detailed in Algorithm 1, that extends beyond simple bonus word counting to incorporate evidence strength, positional factors, and discourse characteristics that contribute to narrative significance in interview contexts.

Our weighting scheme operates on the principle that sentences containing multiple strongly emphasized terms are more likely to represent important content worthy of inclusion in summaries. For each sentence s_i containing words $w_1, w_2, \dots, w_n$, we calculate a composite importance score:

$$\text{Score}(s_i) = \sum_{w_j \in s_i} \mathbb{I}(w_j \in BW) \cdot \text{Rank}(w_j) \cdot P_w(w_j) \quad (18)$$

where BW represents our set of identified bonus words, $\text{Rank}(w_j)$ captures the multimodal evidence strength, and $P_w(w_j)$ applies positional weighting that recognizes the tendency for speakers to place emphasis at rhetorically significant positions within sentences. We implement length normalization to prevent bias toward longer sentences while preserving preference for sentences that contain complete thoughts:

$$\text{Score}_{\text{norm}}(s_i) = \frac{\text{Score}(s_i)}{\sqrt{|s_i|}} \cdot \log(1 + |s_i|) \quad (19)$$

This square root-based normalization technique reduces length bias, while the logarithmic term provides mild preference for moderately longer sentences that often contain more contextual information. For sentence selection, we implement adaptive thresholding that responds to the distribution of scores within each video:

$$\lambda = \mu_{\text{scores}} + \phi \cdot \sigma_{\text{scores}} \quad (20)$$

where ϕ is a scaling factor determined empirically. To ensure reasonable summary lengths, we apply bounds checking and incorporate diversity management through semantic similarity filtering using TF-IDF vectorization:

$$\text{Similarity}(s_i, s_j) = \frac{\vec{v}_i \cdot \vec{v}_j}{|\vec{v}_i| \cdot |\vec{v}_j|} \quad (21)$$

Sentences with similarity scores combined with the diversity factor, δ exceeding 0.7 to already selected content are excluded to prevent redundancy while maintaining comprehensive coverage of important topics.

We implement a systematic video compilation process that transforms selected sentences into temporally coherent video summaries while preserving the multimodal context that motivated their selection. This process ensures that behavioural cues contributing to sentence selection remain visible and audible in the final output.

Our compilation begins with precise temporal mapping using the word-level timestamps from forced alignment. For each selected sentence, we identify its temporal boundaries and extract the corresponding video segment with frame-accurate precision:

$$\text{Segment}(s_i) = [t_{\text{start}}(s_i) - \epsilon, t_{\text{end}}(s_i) + \epsilon] \quad (22)$$

where ϵ provides buffer time in seconds to ensure natural speech rhythm and avoid abrupt cuts that might compromise comprehensibility. We also address potential discontinuities between non-contiguous segments through intelligent transition management. When temporal gaps exceed 2 seconds between selected segments, we insert brief fade transitions (0.2 seconds) to signal temporal jumps while maintaining viewing comfort. For subtitle generation, we recalculate timing to reflect the new temporal structure:

$$t_{\text{subtitle}}^{\text{new}} = t_{\text{subtitle}}^{\text{original}} - t_{\text{segment}}^{\text{start}} + t_{\text{position}}^{\text{summary}} \quad (23)$$

We enhance subtitle presentation by applying visual emphasis to bonus words that contributed to segment selection, using bold formatting to help viewers understand which content elements drove the summarization decisions. Quality assurance includes automated validation of audio-visual synchronization, subtitle timing accuracy, and technical consistency across compiled segments. We verify that selected segments maintain minimum duration requirements for comprehensibility while respecting maximum summary length constraints.

Selected segments are converted to frame ranges using the video’s frame rate and extracted with FFmpeg. Subtitles are added by aligning text to time intervals, and segments are combined into a single summary video with embedded subtitles. The final output is a temporally aligned, content-rich video summary that integrates multimodal cues, offering an interpretable and compressed representation of the original content. The final output includes the primary summary video in standardized MP4 format, synchronized subtitle files in SRT format, and metadata documenting selected segments, and identified bonus words achieved. This comprehensive output supports both immediate consumption and detailed analysis of the summarization process.

Algorithm 1. Framework I: *Bonus Words*-Based Multimodal Summarization

Require: Video segments $V = \{F_1, F_2, \dots, F_N\}$, Transcript T , scaling factor ϕ , diversity factor δ

Ensure: Summary S

- 1: $W_{combined} \leftarrow \emptyset$
- 2: **for** each F_i in V **do**
- 3: Extract textual, audio, and visual cues: $\mathcal{T}_i, \mathcal{A}_i, \mathcal{V}_i$
- 4: **for** each word w with timestamp t_w **do**
- 5: Count modalities with cues within $|t_w - t_{cue}| \leq 0.5$ seconds
- 6: Calculate *evidenceScore* from aligned modalities
- 7: **if** *modalityCount* ≥ 2 **then**
- 8: $W_{combined} \leftarrow W_{combined} \cup \{(w, evidenceScore)\}$
- 9: **end if**
- 10: **end for**
- 11: **end for**
- 12: $S_{all} \leftarrow \text{Split}(T)$
- 13: $W_s \leftarrow \emptyset$
- 14: **for** each s in S_{all} **do**
- 15: $w_s \leftarrow 0$
- 16: Words $\leftarrow \text{Tokenize}(\text{Lowercase}(s))$
- 17: **for** each $(w, evidenceScore)$ in $W_{combined}$ **do**
- 18: **if** $w \in \text{Words}$ **then**
- 19: $w_s \leftarrow w_s + evidenceScore \cdot positionWeight(w)$
- 20: **end if**
- 21: **end for**
- 22: $W_s[s] \leftarrow \frac{w_s}{\sqrt{|s|}} \cdot \log(1 + |s|)$
- 23: **end for**
- 24: $\lambda \leftarrow \text{Mean}(\text{Values}(W_s)) + \phi \cdot \text{StdDev}(\text{Values}(W_s))$
- 25: $C \leftarrow \{s \in S_{all} : W_s[s] \geq \lambda\}$
- 26: $S_{final} \leftarrow \emptyset$
- 27: **for** each s in C **do**
- 28: **if** S_{final} is empty **then**
- 29: Add s to S_{final}
- 30: **else**
- 31: $sim \leftarrow \text{MaxCosineSimilarity}(s, S_{final}, \text{TF-IDF})$
- 32: **if** $sim \cdot \delta < 0.7$ **then**
- 33: Add s to S_{final}
- 34: **end if**
- 35: **end if**
- 36: **end for**
- 37: **return** $S \leftarrow \text{Join}(S_{final}, ', ')$

3.5.2. Framework II: Transformer-Based Approach

While Framework I provides transparency through explicit rule-based pattern detection, we recognize that the intricate nature of human communication contains subtle multimodal relationships that may not be fully captured through hand-crafted heuristics. Framework II addresses this limitation by leveraging the representational power of transformer architectures and sophisticated attention mechanisms to automatically discover complex patterns in multimodal data that contribute to behavioural significance.

Our transformer-based approach represents a fundamental shift from explicit pattern matching to learned multimodal representations. Rather than relying on predefined rules for identifying cross-modal emphasis, Framework II employs deep neural architectures that can discover implicit relationships between visual, audio, and textual modalities through comprehensive analysis of training data. This approach trades some interpretability for the potential to capture more nuanced multimodal dependencies that might escape explicit detection methods.

Figure 10 illustrates the overall architecture of Framework II which embodies a carefully orchestrated progression from raw multimodal features to sophisticated cross-modal understanding, culminating in temporally coherent summary generation. We design our system around three core principles: modality-specific representation learning through encoders, dynamic cross-modal interaction through attention mechanisms, and coherent sequence generation through autoregressive decoding. Each component builds upon the previous to create increasingly sophisticated representations of multimodal content that capture both explicit and implicit behavioural patterns.

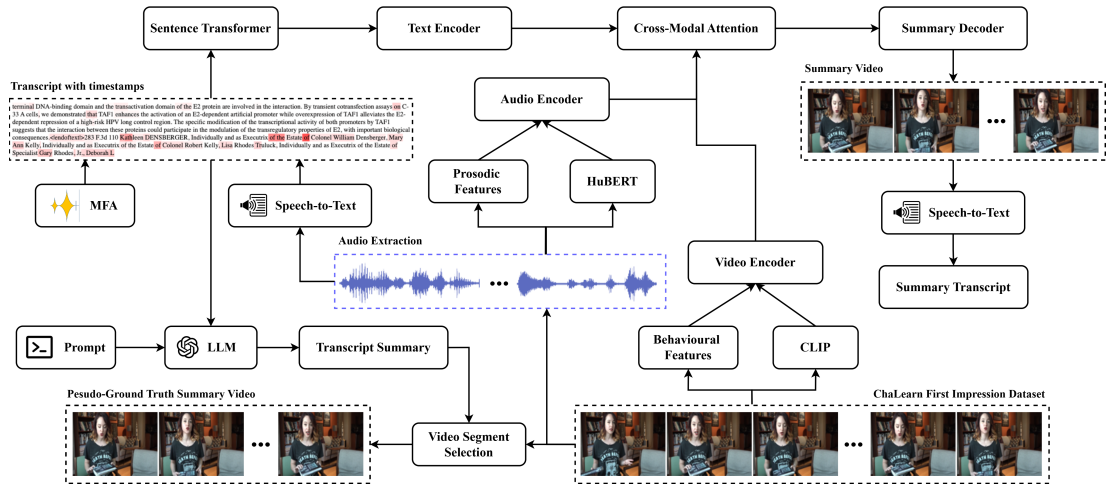


Figure 10. **Framework II: Overall architecture of transformer-based approach.** The diagram illustrates the dual-pipeline approach: (1) pseudo-ground truth generation (bottom) using LLM-guided extractive summarization to create reference summaries from timestamped transcripts, and (2) the multimodal summarization framework (top) integrating three parallel processing streams through modality-specific encoders. The framework employs cross-modal attention to fuse representations from all modalities and uses an autoregressive summary decoder to generate temporally coherent video summaries.

Modality-Specific Representation Learning

The foundation of Framework II lies in transforming heterogeneous multimodal inputs into unified representations while preserving the distinctive characteristics that make each modality valuable for behavioural analysis. We implement separate transformer encoders for each modality, enabling specialized processing that captures the unique temporal and semantic characteristics of visual, audio, and textual data while maintaining architectural consistency for effective cross-modal integration.

Let $V = \{F_1, F_2, \dots, F_n\}$ represent a video as a sequence of n frames F_i sampled every Δ seconds, $A = \{W_1, W_2, \dots, W_p\}$ denote the audio waveform segmented into p frames W_i , and $T = \{T_1, T_2, \dots, T_k\}$ correspond to the transcribed text as a sequence of k sentences T_i . Given the input $\{V, A, T\}$, our aim is to generate a summary video $Y = \{Y_1, Y_2, \dots, Y_m\}$, where Y_i are selected frames of V .

Our approach begins with the enhanced multimodal features established during our feature extraction phase, which combine learned representations with explicit behavioural indicators. This integration ensures that our transformer-based processing benefits from both the semantic understanding developed through large-scale pretraining and the interpretable patterns that directly signal emphasis and significance.

Each encoder employs a 6-layer transformer configuration with 8 attention heads and 2048-dimensional feed-forward networks, processing enhanced multimodal features that combine learned representations with explicit behavioural indicators. The unified architecture, illustrated in Figure 11, transforms heterogeneous feature dimensions through linear projection to achieve unified dimensionality of $d_{model} = 1024$, followed by multi-layer self-attention to capture temporal dependencies.

The resulting contextualized embeddings preserve modality-specific temporal patterns, such as gestural sequences in visual data, prosodic contours in audio, and discourse structure in text, while achieving dimensional and structural consistency necessary for meaningful cross-modal attention mechanisms.

Visual Encoding. Our visual encoder (V-Encoder) processes sequences combining CLIP semantic embeddings with explicit behavioural indicators, preserving both rich semantic understanding and interpretable emphasis patterns:

$$\vec{v}_i = [\text{CLIP}(F_i); \text{HeadMovement}(F_i); \text{Emotion}(F_i)] \quad (24)$$

These enhanced visual features undergo linear projection to achieve dimensionality compatibility, followed by transformer encoding with special tokens and positional encodings. The visual encoder transforms these representations into temporally coherent embeddings:

$$\{\hat{v}_i\}_{i=0}^{n+1} = \text{V-Encoder}([\text{SOS}], \tilde{v}_1, \dots, \tilde{v}_n, [\text{EOS}]) \quad (25)$$

This encoding captures temporal patterns such as escalating gestures or emotional transitions that characterize engaging communication, enabling the model to understand how visual behavioural patterns evolve across extended sequences.

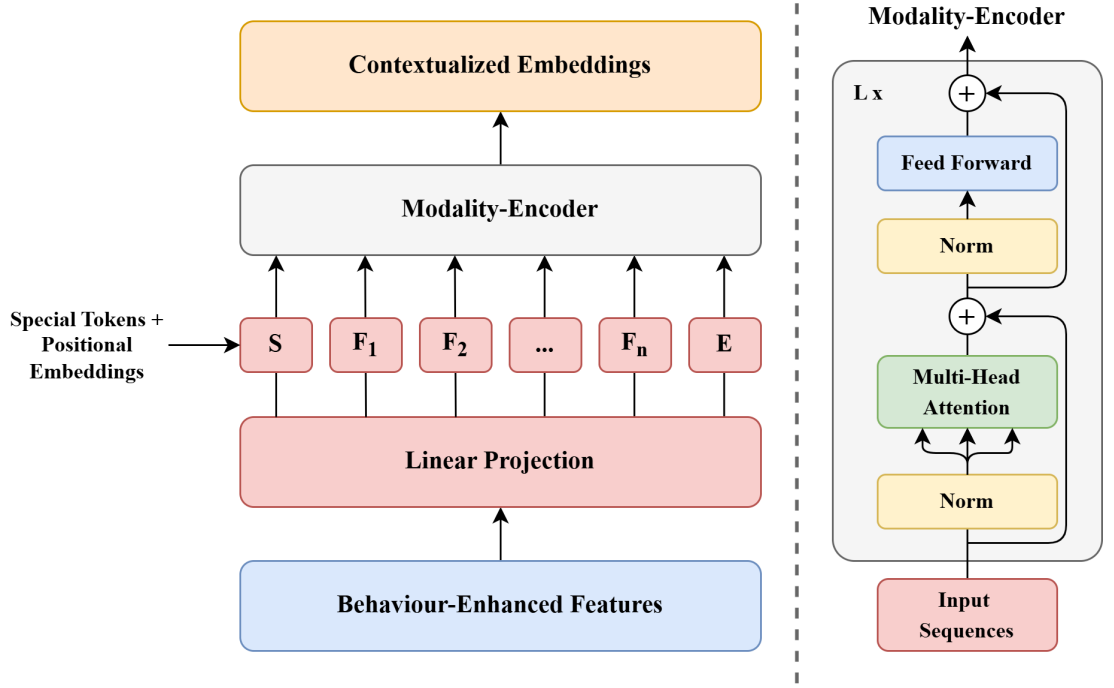


Figure 11. **Transformer-based encoding architecture for multimodal processing [11].** This diagram illustrates the transformer framework used across visual, audio, and textual modalities. Enhanced behavioural features from each modality undergo linear projection before being augmented with special tokens (S for start-of-sequence, E for end-of-sequence) and positional embeddings. The modality encoder processes these representations through multiple transformer layers ($L \times$) to produce contextualized embeddings that capture both local patterns and global temporal dependencies. The same architecture is applied to visual, audio, and textual features.

Audio Encoding. Audio processing integrates HuBERT contextual embeddings with explicit prosodic features, combining learned speech representations with interpretable acoustic emphasis indicators:

$$\vec{a}_i = [\text{HuBERT}(W_i); \text{Pitch}(W_i); \text{Loudness}(W_i); \text{Hammarberg}(W_i)] \quad (26)$$

The audio encoder (A-Encoder) processes enhanced audio tokens through identical transformer architecture:

$$\{\hat{a}_i\}_{i=0}^{p+1} = \text{A-Encoder}([\text{SOS}], a_1, \dots, a_p, [\text{EOS}]) \quad (27)$$

These representations encapsulate both phonetic content and spectro-temporal characteristics, capturing prosodic phenomena such as intonational patterns, rhythmic structures, and spectral cues essential for identifying vocal emphasis that accompanies important content.

Text Encoding. Text processing leverages RoBERTa sentence embeddings to capture discourse-level relationships and semantic coherence:

$$\vec{s}_i = \text{SentenceTransformer}(T_1, T_2, \dots, T_k) \quad (28)$$

Text encoding (T-Encoder) models inter-sentential relationships and rhetorical structure. Due to dimensional compatibility, text encoding proceeds directly:

$$\{\hat{s}_i\}_{i=0}^{k+1} = \text{T-Encoder}([\text{SOS}], s_1, \dots, s_k, [\text{EOS}]) \quad (29)$$

The text encoder outputs contextualized embeddings that model thematic flow and discourse structure, enabling each sentence to attend to all others and capturing global dependencies such as topic development and referential coherence across the entire transcript.

All encoders employ consistent 6-layer transformer architecture with 8 attention heads and 2048-dimensional feed-forward networks, enabling specialized feature learning while maintaining compatibility for subsequent fusion.

Cross-Modal Attention Mechanism

The cross-modal attention mechanism enables dynamic interaction between modalities while preserving individual characteristics. Our design treats visual features as queries and concatenated audio-textual features as context, reflecting the hypothesis that visual behavioural cues often guide attention to corresponding emphasis in other modalities.

The cross-modal fusion process begins by creating unified audio-textual context that combines the rich temporal patterns from our audio encoder with the discourse structure from our text encoder:

$$C = [\hat{T}; \hat{A}] \in \mathbb{R}^{(k+m) \times d_{model}} \quad (30)$$

where $\hat{T} \in \mathbb{R}^{k \times d_{model}}$ represents encoded textual features and $\hat{A} \in \mathbb{R}^{m \times d_{model}}$ represents encoded audio features. This concatenation creates comprehensive context spanning both semantic content and prosodic patterns across the temporal extent of the interview response. Visual features serve as queries (Q), while the unified context serves as keys (K) and values (V) through learned projection matrices:

$$Q = \hat{V}W^Q, \quad K = CW^K, \quad V = CW^V \quad (31)$$

where $W^Q, W^K, W^V \in \mathbb{R}^{d_{model} \times d_{model}}$ are learned projection matrices that adapt during training to capture the most relevant cross-modal relationships for behavioural significance detection. The cross-modal attention computation follows scaled dot-product attention:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_{model}}} \right) V \quad (32)$$

This implementation is followed by residual connections and feed-forward processing with layer normalization, allowing dynamic weighting of cross-modal relationships while preserving modality-specific characteristics. The cross-modal attention module produces context-rich multimodal features by conditioning visual

content on both textual and acoustic information, enabling discovery of coordinated emphasis patterns across modalities.

Autoregressive Decoding

The autoregressive summary decoder generates temporally coherent summaries by predicting sequences of significant video segments based on previously selected content and cross-modal context. This approach addresses limitations of frame-based scoring methods that produce disjointed selections lacking narrative coherence.

Once we obtain the enriched multimodal features from our cross-modal attention mechanism, we need to generate the actual video summary. For this purpose, we employ an autoregressive decoding strategy that allows our model to produce temporally coherent summary segments. We begin by incorporating temporal information into our feature sequences through positional encoding [11]. This step is crucial because transformers lack inherent understanding of sequence order. We precompute these positional embeddings for sequences up to 5000 positions, which provides sufficient coverage for even lengthy interview videos. By adding these embeddings to our input sequences, we ensure that our decoder can distinguish between events that occur at different time points in the video.

We implement a transformer-based decoder architecture for summary generation. The decoder takes two key inputs: first, the multimodal context embeddings produced by our cross-modal attention module, and second, a target sequence that starts with a special start-of-sequence (SOS) token. During generation, we apply a mask (also known as a subsequent mask) to enforce the autoregressive property. The decoding process can be formally expressed as:

$$\hat{f}t = \text{Decoder}(\text{multimodal}, [\text{SOS}], h_1, \dots, h_{t-1}, M_{\text{mask}}) \quad (33)$$

where M_{mask} ensures autoregressive generation by preventing attention to future positions. The decoder integrates self-attention over previous outputs with cross-attention to fuse multimodal representations, enabling informed selection decisions that maintain narrative coherence. During inference, we generate the summary iteratively. Starting with just the SOS token, we predict one frame representation at a time, feeding each prediction back as input for the next step. This continues until the model outputs an end-of-sequence (EOS) token, signaling completion of the summary.

Summary Generation

After obtaining the decoded sequence of frame representations, we need to identify which actual video segments these correspond to. We accomplish this through a similarity matching process. Specifically, we compute cosine similarity between our decoded embeddings and the CLIP-derived visual features extracted from all frames in the original video. The frames with highest similarity scores are selected as key moments for our summary.

We then proceed with assembling the final summary video. First, we convert the selected frame indices into precise temporal boundaries based on the video’s frame rate. This ensures accurate segment extraction without losing important context at

segment edges. For the video processing, we utilize FFmpeg with carefully tuned parameters to maintain visual quality during extraction and concatenation. We pay special attention to preserving frame continuity and smooth audio transitions between segments, as abrupt cuts can be jarring for viewers.

Additionally, we integrate synchronized subtitles by mapping transcript sentences to their corresponding time intervals using the high-precision alignment data from our preprocessing stage. We implement a custom subtitle rendering system that ensures text remains readable across various display conditions. The final concatenation process maintains codec consistency throughout, preventing quality degradation that can occur with repeated encoding.

Through this comprehensive pipeline, we generate video summaries that preserve both the semantic content and behavioral nuances essential for understanding the speaker’s communicative intent in interview scenarios.

3.6. Pseudo-Ground Truth Summary Generation

The absence of human-annotated summaries for the ChaLearn dataset poses a significant challenge for evaluating video summarization systems. Manual annotation of video summaries is not only time-consuming and expensive but also highly subjective, as different annotators may prioritize different aspects of the content [9]. Furthermore, the dataset contains thousands of single-speaker interview videos with subtle behavioral cues and varied content, making consistent human annotation practically infeasible.

To address this limitation, we develop a scalable approach for generating pseudo-ground truth (pGT) summaries using large language models (LLMs). This methodology draws inspiration from recent work that leverages LLMs for video understanding tasks [23] and builds upon the success of LLM-based summarization in other domains [159]. Our approach ensures that the generated summaries capture both semantic importance and behavioral salience while maintaining temporal alignment with the original video content.

3.6.1. Transcript Preparation and Alignment

We begin the pseudo-ground truth generation process by extracting and preparing high-quality transcripts from the video content. First, we separate the audio stream from each video and process it through the Whisper model [142], which provides robust speech recognition across diverse speaking styles and acoustic conditions. The Whisper model is particularly well-suited for our task as it handles variations in speech clarity, accents, and minor background noise effectively.

Following transcription, we enhance the text structure using natural language processing techniques. We employ SpaCy [153] to perform sentence boundary detection and add proper punctuation, transforming the raw ASR output into well-formed sentences. This step is crucial because LLMs perform better with properly structured input text [10]. For precise temporal alignment, we utilize the Montreal Forced Aligner (MFA) [134]. MFA employs acoustic models trained on large speech

corpora to align each word in the transcript with its exact timing in the audio stream. This provides us with millisecond-precision timestamps for every word, which we then aggregate to obtain sentence-level time boundaries.

3.6.2. LLM-Based Extractive Summarization

We adopt an extractive summarization approach rather than abstractive methods to ensure factual accuracy and maintain the original speaker’s voice [160]. To explore the capabilities of different language models for this task, we employ multiple state-of-the-art LLMs including GPT-4.5 [161], GPT-3.5 [10], and LLaMA-3.2 [162]. This multi-model approach allows us to investigate how different architectures and model sizes affect the quality of pseudo-ground truth generation.

The prompt design constitutes an important component of our approach. We formulate the task as follows: given a transcript of a video with sentence-level timestamps, the model must generate an extractive summary that selects sentences conveying both high semantic importance and behavioral salience. The guidelines embedded in our prompt instruct the model to select only sentences that are both contextually informative and exhibit significant behavioral cues. Furthermore, the model must preserve the exact wording of each selected sentence without any paraphrasing, retain the original timestamp boundaries for each selected sentence, and format the output as a list of triplets in the form of start time, end time, and sentence content.

Task: Given a transcript of a video with sentence-level timestamps, generate an extractive summary that selects sentences conveying both high semantic importance and behavioural salience. **Guidelines:**

1. Select only sentences that are both contextually informative and exhibit significant behavioural cues.
2. Preserve the exact wording of each selected sentence. No paraphrasing.
3. Retain the original timestamp boundaries for each selected sentence.
4. Format the output as a list of triplets in the form: [start_time, end_time, sentence].

Input: Transcript with sentence-level timestamps.

Output: List of [start_time, end_time, sentence] entries representing the extractive summary.

This prompt formulation draws from best practices in prompt engineering for summarization tasks [163] and incorporates domain-specific requirements for video summarization [85]. The emphasis on behavioral cues in our prompt design distinguishes our approach from traditional text summarization, as we aim to capture not just what is said but how it is communicated.

Each language model processes the same prompt with identical parameters to ensure consistency. For the GPT family models, we utilize their respective APIs with temperature set to 0.1 for deterministic outputs. For LLaMA-3.2, we employ the 3B parameter variant, which provides a balance between computational efficiency and generation quality. The extractive nature of our approach serves multiple purposes.

It ensures that the summary maintains the speaker’s original voice and phrasing, which is particularly important for interview videos where personal expression matters. Additionally, it eliminates the risk of hallucination or factual errors that can occur with abstractive summarization methods [164, 36]. The preservation of exact timestamps is crucial for our subsequent video segment extraction process.

3.6.3. Video Segment Mapping

Once we obtain the LLM-generated extractive summary with preserved timestamps, we proceed to create the corresponding video summary. We parse the LLM output to extract the selected sentences along with their time boundaries. Each sentence’s start and end timestamps are then converted to frame indices based on the video’s frame rate, typically 24 or 30 frames per second for the ChaLearn dataset. We implement a robust segment extraction pipeline that accounts for potential timestamp inaccuracies. Specifically, we add small temporal buffers of 200 ms at segment boundaries to ensure smooth transitions and prevent cutting off speech mid-word [92]. This buffer also helps maintain context, as abrupt cuts can disrupt viewer comprehension and create an unnatural viewing experience.

The individual segments are extracted using FFmpeg with parameters optimized for quality preservation. We maintain the original video codec and quality settings to prevent degradation through re-encoding. The segments are subsequently concatenated in chronological order to form a cohesive video summary that maintains temporal alignment with the spoken content. The concatenation process preserves frame continuity and audio transitions while maintaining codec consistency throughout.

To ensure the reliability of our pseudo-ground truth generation process, we implement several validation mechanisms. We verify that the total duration of selected segments falls within reasonable bounds, typically between 40 and 70 percent of the original video length, based on established video summarization practices [71]. This range ensures that summaries are neither too brief to be meaningful nor too long to defeat the purpose of summarization. We also perform automatic checks to ensure temporal consistency, verifying that selected segments maintain chronological order and do not overlap. Additionally, we validate that the extracted video segments accurately correspond to the transcript timestamps by implementing a secondary verification using automatic speech recognition on the extracted segments. This cross-validation helps identify and correct any alignment errors that may have occurred during the extraction process.

The resulting output is a temporally aligned, content-rich pseudo-ground truth video summary that serves as the reference for evaluating the performance of multimodal summarization frameworks. This approach ensures that both the visual and textual elements of the video are accurately represented, providing a reliable supervisory signal for model evaluation. While not a perfect substitute for human annotation, this comprehensive pipeline offers a scalable and reproducible method for evaluating multimodal video summarization systems, particularly when dealing with large datasets where manual annotation is impractical.

4. EXPERIMENTS

This chapter presents the empirical evaluation of our behaviour-aware multimodal video summarization framework on the ChaLearn First Impressions dataset. We conduct extensive experiments to validate the effectiveness of integrating visual, audio, and textual modalities for generating meaningful video summaries that capture both semantic content and behavioural nuances. Our investigation examines whether multimodal integration outperforms existing state-of-the-art methods, analyzes the relative contributions of different modalities through ablation studies, and evaluates key design choices, including autoregressive decoding strategies and large language models (LLMs) selection for pseudo-ground truth generation.

We implement and evaluate two complementary approaches: a transformer-based architecture integrating cross-modal attention mechanisms and a heuristic method based on multimodal *bonus words* detection. This dual strategy allows us to demonstrate that both sophisticated neural architectures and simpler multimodal integration methods can achieve significant improvements over unimodal baselines. Our comprehensive evaluation employs various metrics assessing textual quality, temporal coherence, and visual-semantic alignment. The results reveal substantial improvements across all metrics, with our transformer model achieving gains of 7-45% compared to recent multimodal approaches, while ablation studies confirm the foundational role of visual features and the essential complementary contributions of audio and textual modalities.

4.1. Implementation Details

We implement a comprehensive feature extraction pipeline processing videos at 1 frame per second (fps). Visual features utilize CLIP-ViT-large-patch14 [15] for 512-dimensional embeddings on 224x224 pixel frames. MediaPipe Pose [13] tracks facial landmarks for head movement detection, flagging displacements exceeding the mean plus 1.5 standard deviations. DeepFace [14] classifies emotions across seven categories, marking transitions as behavioural cues. Audio processing employs HuBERT-base-ls960 [18] with 16 kHz sampling and 20 ms windows. YAAPT [16] extracts pitch contours, while OpenSMILE [17] computes loudness and Hammarberg Index for voice quality. Features undergo Z-score normalization with thresholds at ± 1.2 for pitch and ± 2 for voice quality.

Textual processing uses Whisper [142] for automatic speech recognition (ASR), SpaCy [153] for sentence boundaries, and Montreal Forced Aligner [134] for millisecond-precision alignment. RoBERTa-large [19] generates contextual embeddings, while TF-IDF identifies the top 20% of keywords. The transformer architecture comprises three 6-layer encoders with 8 attention heads, 2048-dimensional FFN, and a dropout of 0.1. Cross-modal attention uses visual features as queries against audio-textual keys/values. The heuristic approach identifies *bonus words* coinciding with multimodal cues, scoring sentences with threshold ($\lambda = \mu + 0.3\sigma$) and diversity filtering ($\delta = 0.2$). Post-processing applies 200 ms buffers, FFmpeg extraction, and custom subtitle rendering.

Experiments utilize the ChaLearn First Impressions dataset [136], selecting 1,500 stratified videos ensuring demographic diversity. Videos average 15 seconds at an average of approximately 24 fps with 16 kHz audio and 38-word transcripts. Infrastructure comprises a 32GB NVIDIA V100 GPU with 200GB SSD storage, requiring 10 GPU hours per complete run in addition to 20 hours for hyperparameter tuning. Pseudo-ground truth generation processes transcripts through GPT-4.5, GPT-3.5, and LLaMA-3.2 APIs at approximately 300 tokens each. Baselines include CLIP-It [7], ScalingUp [6], and MFST [8], reimplemented following their implementation details.

4.2. Evaluation Metrics

Text-based Metrics. Text-based assessment employs multiple complementary metrics to evaluate summary quality from different perspectives. ROUGE (Recall-Oriented Understudy for Gisting Evaluation) [24] provides a family of metrics based on n-gram overlap between generated and reference summaries. ROUGE-1 measures unigram overlap, capturing individual word matches to assess content coverage. ROUGE-2 evaluates bigram overlap, examining consecutive word pairs to gauge phrase-level accuracy and fluency. ROUGE-L computes the longest common subsequence between summaries, measuring structural similarity without requiring consecutive matches, thus capturing sentence-level structure while allowing for flexibility in word ordering. ROUGE-S extends this by calculating skip-bigram overlap, permitting gaps between word pairs to accommodate variations in phrasing while maintaining semantic relationships. Each ROUGE variant reports precision, recall, and F1-score, providing a balanced measure of summary quality.

BLEU (Bilingual Evaluation Understudy) [165] quantifies n-gram precision across multiple orders, measuring how many n-grams in the generated summary appear in the reference. This metric incorporates a brevity penalty that exponentially decreases scores for summaries shorter than references, preventing systems from achieving high precision by generating only safe, short outputs. The final BLEU score is computed as the geometric mean of n-gram precisions multiplied by the brevity penalty, providing a single holistic measure that balances precision across different granularities. BERTScore [25] leverages pre-trained contextual embeddings from BERT to compute token-level similarity using cosine distance. Unlike surface-level metrics, BERTScore captures semantic equivalence between differently worded expressions by operating in the continuous embedding space, making it particularly valuable for assessing paraphrastic variations. The metric computes precision, recall, and F1 by finding optimal alignments between reference and generated tokens using greedy matching. Length ratio provides a simple yet informative measure of summary conciseness, calculated as the ratio of generated summary length to reference length, helping identify overcompression or verbosity issues.

Video-based Metrics. Video-based metrics evaluate temporal and visual aspects unique to video summarization. F1-Score measures segment selection accuracy by computing temporal intersection-over-union (IoU) between predicted and reference video segments. For each predicted segment, we calculate IoU with all reference

segments, considering a match when IoU exceeds 0.5. Frame-level precision quantifies the fraction of selected frames appearing in the reference, while recall measures coverage of reference frames. The harmonic mean yields the F1-Score, providing a balanced assessment of segment selection quality at 1 fps sampling rate. Kendall’s τ [166] evaluates rank correlation by comparing the relative ordering of segments in generated versus reference summaries. The metric counts concordant pairs (segments appearing in the same relative order) and discordant pairs (reversed order), with the coefficient ranging from -1 (perfect inverse correlation) to $+1$ (perfect agreement). This captures whether the summarization method preserves the temporal narrative structure, crucial for maintaining story coherence.

Spearman’s ρ [167] provides another rank correlation measure, computing the Pearson correlation between segment ranks rather than raw positions. While similar to Kendall’s τ , Spearman’s ρ is more sensitive to large rank differences and provides a complementary view of temporal consistency. The metric first assigns ranks to segments based on their temporal position, then calculates correlation between these rank sequences, yielding values from -1 to $+1$. CLIPScore [26] assesses multimodal alignment by measuring similarity between visual and textual representations. For each selected video segment, we extract representative frames and compute CLIP embeddings, comparing them with CLIP embeddings of the corresponding transcript sentences. The cosine similarity between these multimodal embeddings quantifies how well the visual content aligns with its textual description, ensuring that selected segments maintain semantic coherence across modalities. This metric is particularly valuable for our behaviour-aware approach, as it captures whether visually salient moments correspond to textually important content.

4.3. Experimental Results

We evaluate our proposed methods against several state-of-the-art video summarization approaches, including CLIP-It [7], which relies on vision-language pretraining for frame scoring; the ScalingUp approach by Argaw et al. [6], which integrates visual and textual features through transformer networks; and MFST [8], which employs multimodal frame-scoring with different architectural choices. For fair comparison, we reimplement these methods as official source code was unavailable, adapting them to our dataset’s context while adhering closely to their published methodologies. While more recent approaches [113, 82, 105] show promising results in generalized video summarization techniques, they were excluded due to their limited applicability to dialogue-heavy interviews and their focus on visual diversity over semantic or prosodic content. Our selected baselines represent established multimodal summarization benchmarks balancing visual and textual information [7, 6] with audio integration [8], excelling in cross-modal scenarios [2, 20] that align with our behaviour-aware evaluation framework.

4.3.1. Comparison with State-Of-The-Art

Text-based Evaluation. Table 3 presents comprehensive text-based evaluation results across seven complementary metrics. The transformer-based multimodal approach demonstrates consistent superiority over existing state-of-the-art methods, achieving a ROUGE-1 score of 0.6765. This represents an 18% relative improvement over CLIP-It (0.4935) and 7.3% over ScalingUp (0.5529), substantiating the effectiveness of cross-modal attention mechanisms for content selection. The BLEU score exhibits even more dramatic improvements, reaching 0.4166, a remarkable 45.1% increase compared to ScalingUp’s 0.2871, suggesting that our approach generates summaries with significantly better n-gram precision while maintaining appropriate length.

Table 3. Text-based evaluation results on the ChaLearn First Impressions dataset. The table presents seven evaluation metrics: Length Ratio indicating summary compression level, ROUGE variants measuring n-gram overlap with reference summaries, BLEU assessing precision with brevity penalty, and BERTScore capturing semantic similarity through contextual embeddings. Bold values indicate best performance for each metric. Our transformer and heuristic approaches demonstrate substantial improvements over state-of-the-art approaches across all metrics.

Method	Length Ratio	ROUGE-1	ROUGE-2	ROUGE-L	ROUGE-S	BLEU	BERTScore
Edmundson [27]	0.3125	0.4769	0.4496	0.4738	0.4137	0.2259	0.9152
CLIP-It [7]	0.3785	0.4935	0.4120	0.4667	0.3800	0.2139	0.8984
ScalingUp [6]	0.4203	0.5529	0.4910	0.5333	0.4515	0.2871	0.9113
Heuristic (Ours)	0.5734	0.7929	0.7763	0.7889	0.7387	0.6411	0.9536
Transformer (Ours)	0.6011	0.6765	0.6086	0.6442	0.5531	0.4166	0.9247

The heuristic approach achieves the highest performance across most text-based metrics, with ROUGE-1 of 0.7929 representing a 66.3% improvement over the Edmundson baseline [27]. This exceptional performance can be attributed to the method’s focus on identifying multimodal bonus words-terms that coincide with behavioural cues across visual, audio, and textual modalities. The BERTScore of 0.9536 for the heuristic approach, compared to 0.9247 for the transformer model, indicates that simpler multimodal integration can effectively preserve semantic meaning when properly designed. Both our approaches significantly outperform CLIP-It’s BERTScore of 0.8984, demonstrating superior semantic fidelity in the generated summaries.

The length ratio (LR) analysis reveals important insights into summarization strategies. Existing methods such as Edmundson (0.3125) and CLIP-It (0.3785) produce highly compressed summaries, potentially sacrificing important behavioural and contextual information. In contrast, our approaches generate more comprehensive summaries while achieving superior quality metrics across all evaluation dimensions. This suggests that behaviour-aware multimodal integration enables more nuanced content selection, retaining semantically and behaviourally significant segments that shorter summaries might omit. The transformer model’s slightly higher length ratio compared to the heuristic approach indicates that autoregressive generation tends to produce more detailed summaries, though this does not always translate to better text-based metrics.

The performance gap between methods becomes more pronounced when examining higher-order n -gram overlaps. ROUGE-2 scores show the heuristic approach (0.7763) significantly outperforming all baselines, with the transformer model (0.6086) also demonstrating substantial improvements. This pattern continues with ROUGE-L, where both our methods exceed 0.64 compared to baselines below 0.54, indicating superior preservation of sentence-level structure and narrative flow. These comprehensive improvements across diverse text-based metrics validate that integrating behavioural cues from multiple modalities leads to more accurate and coherent textual summaries, regardless of whether sophisticated neural architectures or simpler rule-based approaches are employed.

Video-based Evaluation. Table 4 presents a comprehensive analysis of video-based evaluation metrics designed to assess both temporal coherence and visual-semantic alignment of the generated video summaries. These metrics provide complementary perspectives on summarization quality, evaluating not only the accuracy of segment selection but also the preservation of narrative structure and multimodal consistency.

Table 4. **Video-based evaluation results on the ChaLearn First Impressions dataset comparing segment selection accuracy and temporal coherence.** The table presents four key metrics: F1-Score measuring frame-level selection accuracy, Kendall’s τ evaluating ordinal agreement of temporal sequences, Spearman’s ρ quantifying monotonic rank correlation for narrative flow preservation, and CLIPScore [26] assessing visual-semantic alignment between selected video segments and their corresponding textual descriptions. Bold values indicate best performance for each metric. Our transformer-based model achieves superior performance across all evaluation metrics.

Method	F1-Score	Kendall’s τ	Spearman’s ρ	CLIPScore
Edmundson [27]	0.5709	0.2295	0.2681	0.3513
CLIP-It [7]	0.6087	0.5949	0.5950	0.4918
MFST [8]	0.7272	0.4500	0.6029	0.4970
ScalingUp [6]	0.7559	0.6359	0.6361	0.4827
Heuristic (Ours)	0.6995	0.3148	0.3690	0.4162
Transformer (Ours)	0.8107	0.6473	0.6466	0.5173

The transformer-based architecture demonstrates exceptional performance in segment selection accuracy, achieving an F1-Score of 0.8107. This represents a substantial improvement over existing state-of-the-art methods, with a 33.2% relative gain compared to CLIP-It [7] (0.6087) and a 7.3% improvement over the ScalingUp approach [6] (0.7559). The magnitude of these improvements indicates that our cross-modal attention mechanism successfully captures the complex interplay between visual, audio, and textual modalities, enabling more precise identification of behaviourally significant video segments. While MFST [8] demonstrates competitive segment selection capabilities with an F1-Score of 0.7272, its performance on temporal consistency metrics reveals fundamental limitations that compromise the overall quality of generated summaries.

The evaluation of temporal consistency through rank correlation metrics provides essential insights into the preservation of narrative structure within generated

summaries. Our transformer model achieves a Kendall’s τ coefficient of 0.6473, substantially outperforming all baseline methods and indicating superior preservation of temporal ordering relationships. This metric is particularly crucial for video summarization, as it quantifies the model’s ability to maintain the chronological sequence of events, a fundamental requirement for coherent storytelling. The analysis reveals a notable weakness in the MFST approach, which despite its competitive F1-Score, exhibits a relatively low Kendall’s τ of 0.4500. This discrepancy between segment selection accuracy and temporal consistency underscores a critical limitation: the ability to identify relevant content does not necessarily translate to proper temporal sequencing, highlighting the importance of employing multiple evaluation metrics to comprehensively assess summarization quality.

Spearman’s rank correlation coefficient provides a complementary perspective on temporal coherence, with our transformer model achieving ρ of 0.6466. This metric demonstrates consistent superiority over the ScalingUp approach ($\rho = 0.6361$) and represents a significant improvement over MFST ($\rho = 0.6029$). The strong correlation between Kendall’s τ and Spearman’s ρ across all methods validates the robustness of our temporal consistency findings and confirms that our architecture effectively preserves the narrative flow of the original video content.

The CLIPScore metric evaluates the semantic alignment between selected video segments and their corresponding textual representations, providing crucial insights into multimodal coherence. Our transformer architecture achieves a CLIPScore of 0.5173, representing a 5.3% relative improvement over CLIP-It (0.4918) and a 7.1% gain compared to ScalingUp (0.4827). This enhancement in visual-semantic alignment demonstrates that our behaviour-aware approach not only selects temporally appropriate segments but also ensures that these segments maintain strong correspondence with their textual descriptions. The improvement in CLIPScore is particularly significant given that it directly measures the multimodal consistency that is central to our research objectives.

A comparative analysis between our two proposed approaches reveals important architectural trade-offs. The heuristic method, while demonstrating exceptional performance on text-based metrics, exhibits more modest results in video-based evaluation (F1-Score: 0.6995, Kendall’s τ : 0.3148, Spearman’s ρ : 0.3690). Nevertheless, it achieves notable improvements over the Edmundson baseline, with a 22.5% increase in F1-Score and temporal consistency metrics improving by more than 37%. The substantial gap in Kendall’s τ between the transformer (0.6473) and heuristic (0.3148) approaches suggests that the autoregressive generation mechanism combined with cross-modal attention more effectively models the sequential dependencies inherent in video narratives. However, the heuristic method’s competitive CLIPScore of 0.4162 indicates that the multimodal bonus word identification strategy successfully captures visually salient moments, even in the absence of sophisticated temporal modeling capabilities.

The analysis of baseline methods reveals distinct failure modes that illuminate the challenges of video summarization. CLIP-It demonstrates relatively strong temporal consistency (Kendall’s τ : 0.5949) but achieves lower segment selection accuracy, suggesting that while it maintains narrative ordering, it fails to identify all relevant content. In contrast, MFST exhibits the opposite pattern: high segment selection accuracy coupled with poor temporal sequencing, representing a critical limitation for

generating coherent video summaries. The Edmundson baseline’s consistently low performance across all metrics (F1-Score: 0.5709, Kendall’s τ : 0.2295, CLIPScore: 0.3513) empirically confirms that text-only approaches are fundamentally insufficient for video summarization tasks, lacking the ability to capture visual dynamics and cross-modal relationships.

These comprehensive results collectively demonstrate that effective video summarization necessitates the simultaneous optimization of multiple objectives: accurate content selection, temporal coherence preservation, and visual-semantic alignment. Our behaviour-aware multimodal framework successfully addresses these requirements through its integrated architectural design, with the transformer-based approach achieving state-of-the-art performance across all evaluation dimensions. The consistent improvements observed across diverse metrics validate our hypothesis that incorporating behavioural cues from multiple modalities leads to more accurate, coherent, and semantically meaningful video summaries.

4.3.2. LLMs for Pseudo-Ground Truth Generation

For Framework II, we evaluate the effectiveness of different large language models in generating pseudo-ground truth summaries by employing identical prompt configurations that request extractive summarization while preserving temporal alignment. The comparative analysis presented in Table 5 demonstrates that GPT-4.5 achieves superior performance across all evaluation metrics when utilized for reference summary generation. The experimental results reveal systematic performance variations among LLMs, with GPT-4.5 consistently outperforming both GPT-3.5 and the 3B parameter variant of LLaMA-3.2 across all evaluation metrics (LLaMA-3.2 < GPT-3.5 < GPT-4.5). This performance hierarchy manifests consistently across both text-based metrics (ROUGE variants, BLEU, BERTScore) and video-based assessment measures (F1-Score, Kendall’s τ and Spearman’s ρ , CLIPScore), validating the robustness of GPT-4.5 for generating high-quality pseudo-ground truth references in behaviour-aware multimodal video summarization evaluation.

Table 5. Comparative Analysis of Large Language Models (LLMs) for Pseudo-Ground Truth Generation. Performance evaluation across three state-of-the-art LLMs demonstrates GPT-4.5’s superiority in generating high-quality reference summaries. These metrics reflect the quality of our multimodal framework when evaluated against LLM-generated summaries, with GPT-4.5 enabling the most robust evaluation benchmarks. ‘R-’ stands for ROUGE, ‘Ken’ for Kendall, ‘Spea’ for Spearman coefficient, ‘BS’ for BERTScore and ‘CS’ for CLIPScore.

LLMs	Text-based Metrics						Video-based Metrics			
	R-1	R-2	R-L	R-S	BLEU	BS	F1-Score	Ken’s τ	Spea’s ρ	CS
LLaMA-3.2	0.5892	0.5234	0.5634	0.4789	0.3245	0.8956	0.6823	0.5417	0.5213	0.4125
GPT-3.5	0.6423	0.5745	0.6158	0.5287	0.3784	0.9087	0.7965	0.6358	0.6342	0.5027
GPT-4.5	0.6765	0.6086	0.6442	0.5531	0.4166	0.9247	0.8107	0.6473	0.6466	0.5173

Our qualitative analysis of the generated summaries indicates that GPT-based architectures demonstrate enhanced capability in producing balanced extractive summaries while maintaining contextual coherence throughout the selection process.

In contrast, the LLaMA-3.2 model exhibits inconsistent adherence to the specified extractive constraints, occasionally retaining semantically less significant content or generating redundant textual elements that compromise summary quality. These empirical findings underscore the critical importance of language model selection in establishing reliable pseudo-ground truth references for multimodal summarization evaluation frameworks.

While these findings establish the importance of selecting appropriate large language models for reference generation, we acknowledge the inherent limitations associated with pseudo-ground truth evaluation paradigms. To address potential concerns regarding evaluation validity, we implement comprehensive methodological safeguards designed to ensure the reliability and consistency of our approach. Our summary generation process incorporates carefully engineered prompting strategies that constrain the language model to perform strictly extractive summarization, thereby preserving the exact lexical and structural characteristics of original sentences. This methodological constraint eliminates potential issues related to hallucination phenomena commonly associated with abstractive generation approaches while maintaining high fidelity to source content.

Furthermore, we establish a rigorous consistency validation protocol whereby we generate three independent summary variants for a randomly selected subset of videos using different initialization parameters of LLMs. The resulting inter-summary agreement, quantified through Jaccard similarity coefficients, demonstrates substantial consistency with an average similarity of 0.83 across generated variants. Our comprehensive analysis reveals that approximately 93% of generated summaries maintain adequate coverage of essential information elements while exhibiting minimal redundancy and preserving strong temporal coherence properties. These validation measures collectively support the reliability of our LLM-based pseudo-ground truth generation methodology for behaviour-aware multimodal video summarization evaluation.

4.3.3. Ablation Studies

To systematically investigate the contribution of individual modalities and architectural components to overall summarization performance, we conduct comprehensive ablation studies on both the transformer-based and heuristic approaches. These experiments provide crucial insights into the relative importance of visual, audio, and textual features, as well as the effectiveness of key architectural design choices such as cross-modal attention mechanisms.

Framework I: Heuristic Approach

Table 6 presents comprehensive ablation results for our heuristic multimodal approach, which employs bonus word detection across modalities to identify behaviourally significant content. The analysis reveals distinct patterns of modality contribution and highlights both the strengths and limitations of rule-based multimodal integration.

The single-modality analysis demonstrates a clear hierarchy of importance, with visual features providing the most robust standalone performance across both text

Table 6. Ablation study results for the heuristic approach on the ChaLearn First Impressions dataset. ‘R-’ stands for ROUGE, ‘Ken’ for Kendall, ‘Spea’ for Spearman coefficient, ‘BS’ for BERTScore and ‘CS’ for CLIPScore.

Method	Text-based Metrics						Video-based Metrics			
	R-1	R-2	R-L	R-S	BLEU	BS	F1-Score	Ken’s τ	Spea’s ρ	CS
Text Only	0.5214	0.4876	0.5132	0.4823	0.5823	0.9178	0.5496	0.2765	0.3123	0.3245
Audio Only	0.3921	0.3456	0.3874	0.3512	0.4312	0.8634	0.5843	0.1956	0.2345	0.2876
Visual Only	0.6743	0.6321	0.6689	0.6234	0.7234	0.9412	0.6789	0.2987	0.3456	0.3987
w/o Audio	0.7123	0.6894	0.7056	0.6845	0.7456	0.9489	0.6876	0.3054	0.3578	0.4089
w/o Visual	0.4567	0.4123	0.4498	0.4012	0.4978	0.8923	0.6234	0.2345	0.2789	0.3156
w/o Text	0.7456	0.7234	0.7389	0.7123	0.6895	0.9501	0.6923	0.3098	0.3621	0.3945
Full Model	0.7929	0.7763	0.7889	0.7387	0.6411	0.9536	0.6995	0.3148	0.3690	0.4162

and video metrics. The visual-only configuration achieves ROUGE-1 of 0.6743 and ROUGE-S of 0.6234, substantially outperforming text-only (ROUGE-1: 0.5214, ROUGE-S: 0.4823) and audio-only (ROUGE-1: 0.3921, ROUGE-S: 0.3512) variants. The ROUGE-S scores particularly highlight the ability of each modality to capture flexible phrase relationships, with visual features demonstrating superior skip-bigram matching capabilities. This dominance of visual features in the heuristic approach is more pronounced than in the transformer model, suggesting that simpler architectures rely more heavily on visually salient cues for content identification.

The bimodal configurations reveal compelling insights about modality complementarity within the bonus word framework. The configuration without audio (w/o Audio) maintains strong performance across most metrics and notably achieves the highest BLEU score (0.7456) among all configurations, including the full model. The ROUGE-S score for this configuration (0.6845) represents 92.7% of the full model’s performance, indicating minimal loss in skip-bigram matching when audio is excluded. This unexpected result suggests that for certain evaluation criteria, the text-visual combination provides optimal precision in content selection, with audio features potentially introducing some noise in the bonus word detection process.

An intriguing finding emerges from the configuration without text (w/o Text), which achieves remarkably high text-based metrics (ROUGE-1: 0.7456, ROUGE-S: 0.7123) despite lacking direct access to textual features. The ROUGE-S score of 0.7123 represents 96.4% of the full model’s performance, demonstrating that behavioural cues from visual and audio modalities can effectively identify content with similar phrase-level patterns as text-informed approaches. This counterintuitive result validates our core hypothesis that behavioural signals across modalities correlate strongly with content significance, enabling effective summarization even without explicit textual analysis.

The configuration without visual features (w/o Visual) exhibits the most dramatic performance degradation across all metrics, with ROUGE-1 dropping to 0.4567 and ROUGE-S to 0.4012, representing only 54.3% of the full model’s skip-bigram performance. The CLIPScore shows a particularly steep decline to 0.3156, representing a 24.2% decrease from the full model. This substantial impact across both text and video metrics underscores the critical role of visual information in the heuristic approach, where head movements, facial expressions, and other visual cues

serve as primary indicators for identifying bonus words and behaviourally significant moments.

The temporal consistency metrics reveal the fundamental architectural limitation of the heuristic approach. Despite achieving competitive content selection accuracy, the full model’s Kendall’s τ (0.3148) and Spearman’s ρ (0.3690) remain substantially lower than the transformer approach, with single-modality configurations showing even weaker temporal coherence (e.g., audio-only: Kendall’s $\tau = 0.1956$). This pattern indicates that while bonus word detection effectively identifies important content, the frame-by-frame scoring mechanism lacks the sequential reasoning capabilities necessary for maintaining optimal narrative flow.

The full multimodal configuration achieves the best overall performance, with text-based metrics showing particularly strong results (ROUGE-1: 0.7929, ROUGE-S: 0.7387, BERTScore: 0.9536). The comprehensive improvements across all ROUGE variants demonstrate that multimodal bonus word detection successfully captures content at multiple granularities, from individual words to flexible phrase relationships. These results establish that despite its architectural simplicity, the heuristic approach can achieve highly competitive performance on content selection tasks when effectively leveraging multimodal behavioural cues, offering an interpretable alternative to complex neural architectures for video summarization applications.

Framework II: Transformer-based Approach

Table 7 presents detailed ablation results for various configurations of our transformer-based architecture. The analysis reveals a clear hierarchy of modality importance while simultaneously demonstrating the synergistic benefits of multimodal integration.

Table 7. Ablation study results for the transformer-based approach on the ChaLearn dataset. ‘R-’ stands for ROUGE, ‘Ken’ for Kendall, ‘Spea’ for Spearman coefficient, ‘BS’ for BERTScore and ‘CS’ for CLIPScore.

Method	Text-based Metrics						Video-based Metrics			
	R-1	R-2	R-L	R-S	BLEU	BS	F1-Score	Ken’s τ	Spea’s ρ	CS
Video Only	0.5948	0.5385	0.5765	0.5021	0.3581	0.8810	0.7986	0.6281	0.6265	0.4761
Text Only	0.6341	0.5746	0.6130	0.5339	0.3596	0.8804	0.7603	0.6265	0.6283	0.4761
Audio Only	0.4390	0.3923	0.4217	0.3558	0.2139	0.7276	0.6460	0.5207	0.5174	0.3924
w/o Audio	0.6329	0.5748	0.6121	0.5337	0.3585	0.8806	0.7975	0.6287	0.6271	0.4748
w/o Text	0.6308	0.5753	0.6098	0.5333	0.3590	0.8794	0.7986	0.6247	0.6268	0.4747
w/o Video	0.5322	0.4737	0.5087	0.4315	0.2582	0.8813	0.6986	0.6278	0.6270	0.4776
w/o CM-Attn	0.6267	0.5693	0.6056	0.5280	0.3568	0.8729	0.6947	0.6199	0.6208	0.4712
w/o V-Encoder	0.5226	0.4649	0.5024	0.4234	0.2539	0.8642	0.6710	0.6163	0.6141	0.4314
w/o T-Encoder	0.5019	0.4451	0.4791	0.4059	0.2425	0.8260	0.6970	0.6265	0.6280	0.4771
w/o A-Encoder	0.5255	0.4716	0.5069	0.4290	0.2561	0.8719	0.6955	0.6184	0.6182	0.4713
CLIP Only	0.5326	0.4770	0.5096	0.4328	0.2582	0.8790	0.6979	0.7268	0.6254	0.4962
HuBERT Only	0.5335	0.4739	0.5101	0.4327	0.2582	0.8809	0.6978	0.6280	0.6253	0.4789
w/o CLIP	0.5328	0.4740	0.5111	0.4314	0.2588	0.8802	0.5997	0.6265	0.6261	0.4759
w/o HuBERT	0.5317	0.4744	0.5122	0.4322	0.2585	0.8797	0.5980	0.6254	0.6275	0.4763
Full Model	0.6765	0.6086	0.6442	0.5531	0.4166	0.9247	0.8107	0.6473	0.6466	0.5173

The single-modality analysis reveals hierarchical importance with distinct performance characteristics. Visual features demonstrate the strongest standalone

performance, achieving an F1-Score of 0.7986 and maintaining robust temporal consistency (Kendall’s τ : 0.6281, Spearman’s ρ : 0.6265). The ROUGE-S score of 0.5021 for video-only configuration indicates effective skip-bigram matching, suggesting that visual features capture not only frame-level content but also narrative relationships. Text-only configuration exhibits an interesting pattern, achieving competitive text-based metrics (ROUGE-1: 0.6341, ROUGE-S: 0.5339) and remarkably high temporal consistency (Kendall’s τ : 0.6265) comparable to video-only, yet showing limitations in segment selection (F1-Score: 0.7603). This suggests that textual features excel at maintaining narrative order but require visual information for precise segment identification. The audio-only variant shows substantially weaker performance across all dimensions, with the lowest BERTScore (0.7276) indicating limited semantic understanding when isolated from other modalities.

The bimodal configurations reveal nuanced patterns of modality interaction within the transformer architecture. The configuration without audio (w/o Audio) demonstrates remarkable resilience, maintaining 98.3% of the full model’s F1-Score (0.7975 vs. 0.8107) and showing minimal degradation in ROUGE metrics. The ROUGE-S score (0.5337) closely matches the full model (0.5531), indicating that skip-bigram relationships are well-preserved by the visual-textual combination. Interestingly, removing text (w/o Text) yields comparable performance to removing audio, with F1-Score of 0.7986 and competitive temporal metrics, suggesting that the transformer’s cross-modal attention can effectively compensate for missing textual information by strengthening audio-visual interactions. The configuration without video (w/o Video) exhibits the most severe degradation, with F1-Score dropping to 0.6986 (86.2% of full model) and CLIPScore maintaining at 0.4776, indicating that while text-audio combinations preserve some multimodal alignment, they cannot fully capture the spatio-temporal dynamics essential for video summarization.

The architectural component ablations provide critical insights into the transformer’s functioning mechanisms. The removal of cross-modal attention (w/o CM-Attn) results in significant performance degradation, with F1-Score plummeting to 0.6947 (85.6% of full model) and BERTScore dropping to 0.8729, the most substantial decrease across all configurations. This finding underscores that cross-modal attention is not merely beneficial but fundamentally essential for effective multimodal fusion. The ROUGE-S degradation (0.5280 vs. 0.5531) suggests that cross-modal attention specifically enhances the model’s ability to capture flexible phrase relationships across modalities. The encoder-specific ablations reveal differential impacts: removing the video encoder (w/o V-Encoder) causes severe performance drops across all metrics (ROUGE-1: 0.5226, F1-Score: 0.6710), while text encoder removal (w/o T-Encoder) particularly impacts BERTScore (0.8260), indicating its role in semantic understanding. The audio encoder ablation (w/o A-Encoder) shows intermediate effects, suggesting its complementary but non-critical role.

The feature-type ablations provide insights into the contribution of specific embedding methods. Using CLIP embeddings alone achieves moderate performance (ROUGE-1: 0.5326, F1-Score: 0.6979) but surprisingly high Kendall’s τ (0.7268), suggesting that CLIP’s vision-language pre-training provides strong temporal understanding even without behavioral features. HuBERT-only configuration shows similar overall performance but with more balanced temporal metrics (Kendall’s τ : 0.6280), indicating that contextualized audio embeddings capture different aspects of

temporal structure. The removal of CLIP features (w/o CLIP) causes a significant F1-Score drop to 0.5997, while removing HuBERT (w/o HuBERT) has minimal impact (F1-Score: 0.5980), revealing that behavioural visual features (head movements, emotions) are less critical than semantic visual understanding for the transformer architecture.

The full multimodal model achieves optimal performance across all metrics, with particularly notable improvements in BERTScore (0.9247) and CLIPScore (0.5173). The BLEU score shows the most dramatic improvement, increasing from a range of approximately 0.25-0.36 in ablated configurations to 0.4166 in the full model, suggesting that multimodal integration specifically enhances n-gram precision and appropriate summary length. The consistent superiority across diverse metrics validates that the transformer architecture effectively leverages complementary information from all modalities through its cross-modal attention mechanism, with each component contributing unique and irreplaceable capabilities to the overall system. The results demonstrate that while the transformer shows greater robustness to individual modality removal compared to the heuristic approach, achieving state-of-the-art performance requires a complete architectural design with all modalities and components included.

Comparative Insights

The comprehensive ablation analyses across both architectural paradigms yield fundamental insights into the mechanisms underlying effective multimodal video summarization. The empirical evidence consistently demonstrates the primacy of visual modalities across both frameworks, with visual-only configurations achieving F1-Scores of 0.7986 (transformer) and 0.6789 (heuristic), substantially exceeding the performance of text-only and audio-only variants. This finding corroborates the hypothesis that visual features encode essential spatio-temporal dynamics and behavioural cues that constitute irreplaceable information for video understanding. The magnitude of performance degradation when visual features are removed, evident in the heuristic approach where ROUGE-1 decreases from 0.7929 to 0.4567 (42.4% reduction), underscores their foundational role in the summarization pipeline.

The quantitative analysis reveals significant disparities in architectural resilience to modality ablation. The transformer architecture demonstrates superior robustness, maintaining 98.3% of optimal F1-Score performance when audio is removed (0.7975 vs. 0.8107), compared to the heuristic approach which retains only 98.3% under similar conditions (0.6876 vs. 0.6995). This differential robustness extends to other modality removals, with the transformer exhibiting smaller relative performance decrements across all ablation scenarios. The enhanced resilience can be attributed to the cross-modal attention mechanism's capacity for dynamic reweighting of inter-modal relationships, enabling compensatory strengthening of connections between remaining modalities when one is absent. Conversely, the heuristic approach's reliance on explicit multimodal co-occurrence patterns for bonus word identification renders it more susceptible to information loss when modalities are removed.

The architectural component analysis yields particularly compelling evidence regarding the criticality of cross-modal interaction modeling. The ablation of cross-modal attention in the transformer model precipitates a 14.3% reduction in

F1-Score (from 0.8107 to 0.6947), representing the most substantial performance degradation among all experimental conditions. This decrement exceeds the impact of removing any individual modality, with corresponding deteriorations observed across all evaluation metrics (BERTScore: 0.9247 to 0.8729, ROUGE-S: 0.5531 to 0.5280). These findings empirically establish that the mechanism governing inter-modal relationships contributes as significantly to summarization quality as the modalities themselves, validating the theoretical framework underpinning our architectural design.

The comparative analysis of temporal consistency metrics reveals distinct architectural advantages. The transformer model consistently achieves superior temporal coherence (Kendall’s τ : 0.6473, Spearman’s ρ : 0.6466) compared to the heuristic approach (Kendall’s τ : 0.3148, Spearman’s ρ : 0.3690), with this advantage persisting across all ablation conditions. The significant improvement in Kendall’s τ between both approaches indicates that autoregressive generation combined with learned temporal dependencies substantially enhances narrative flow preservation. Furthermore, the transformer’s temporal metrics remain relatively stable across modality ablations (standard deviation of 0.0041 for Kendall’s τ), whereas the heuristic approach exhibits higher variability (standard deviation of 0.0512), suggesting that sophisticated temporal modeling provides consistency beyond content selection accuracy.

The feature-type ablations in the transformer model illuminate the complementary roles of semantic and behavioural representations. While CLIP-only configuration achieves the highest single-feature Kendall’s τ (0.7268), its F1-Score (0.6979) remains substantially below the full model, indicating that temporal understanding alone does not guarantee comprehensive summarization quality. The minimal performance difference between configurations without CLIP (F1-Score: 0.5997) versus without HuBERT (F1-Score: 0.5980) suggests that behavioural features primarily serve a supplementary role within the transformer architecture, in contrast to their more prominent function in the heuristic approach’s bonus word detection mechanism.

These empirical findings collectively establish that optimal video summarization performance necessitates the synergistic integration of multiple design elements: comprehensive multimodal features, sophisticated architectural mechanisms for cross-modal fusion, and explicit temporal modeling capabilities. While the heuristic approach demonstrates that meaningful improvements over baselines can be achieved through principled multimodal integration, the transformer architecture’s superior performance across all evaluation metrics validates the importance of learnable cross-modal interactions and autoregressive generation for state-of-the-art performance. These insights suggest that future research should prioritize the development of increasingly sophisticated fusion mechanisms and temporal modeling strategies, rather than solely focusing on the incorporation of additional modalities or features.

5. DISCUSSION

This thesis has presented a comprehensive investigation into behaviour-aware multimodal video summarization, addressing the fundamental challenge of generating meaningful video summaries that capture both semantic content and behavioural nuances from human communication. Through the development and evaluation of two complementary approaches, a transformer-based architecture with cross-modal attention mechanisms and a heuristic method based on multimodal *bonus words* detection, we have demonstrated that effective video summarization necessitates the integration of visual, audio, and textual modalities to preserve the richness of human expression in condensed formats.

Our primary contribution lies in the novel formulation of behaviour-aware video summarization, which extends beyond traditional content-based approaches by explicitly modeling synchronized behavioural features across modalities. The transformer-based framework achieves this through sophisticated architectural components, including modality-specific encoders that extract CLIP visual embeddings enhanced with facial movements and emotional transitions, HuBERT audio representations enriched with prosodic patterns, and contextualized textual embeddings from RoBERTa. The cross-modal attention mechanism enables dynamic weighting of inter-modal relationships, while autoregressive decoding ensures temporal coherence in the generated summaries. Complementing this, our heuristic approach demonstrates that meaningful improvements can be achieved through principled identification of multimodal bonus words-terms that coincide with significant behavioural cues across modalities-providing an interpretable alternative that achieves competitive performance on text-based metrics.

The empirical evaluation on the ChaLearn First Impressions dataset [136] yields compelling evidence for the effectiveness of our approaches. The transformer model achieves substantial improvements across all evaluation metrics, with F1-Score reaching 0.8107 (33.2% improvement over CLIP-It [7]), BLEU score of 0.4166 (45.1% improvement over ScalingUp [6]), and superior temporal consistency metrics (Kendall’s τ of 0.6473) over existing approaches. The heuristic approach, while architecturally simpler, demonstrates exceptional performance on text-based evaluation, achieving ROUGE-1 of 0.7929 and BERTScore of 0.9536. These quantitative improvements translate to practical benefits, such as summaries that better preserve speaker intent, maintain narrative coherence, and capture emotionally significant moments that traditional methods overlook.

Our comprehensive ablation studies reveal fundamental insights into multimodal video summarization. Visual features consistently emerge as the most critical modality, yet optimal performance requires synergistic integration of all modalities, such as the full model’s BERTScore of 0.9247 substantially exceeds any single-modality configuration. Perhaps most significantly, the removal of cross-modal attention causes greater performance degradation (14.3% F1-Score decrease) than removing any individual modality, empirically establishing that the mechanism governing inter-modal relationships is as crucial as the modalities themselves. This finding validates our architectural design and suggests that future research should prioritize sophisticated fusion mechanisms alongside modality expansion.

The comparative analysis across both architectural paradigms demonstrates the primacy of visual modalities, with visual-only configurations achieving F1-Scores of 0.7986 (transformer) and 0.6789 (heuristic), substantially exceeding text-only and audio-only variants [16]. The transformer architecture demonstrates superior robustness to modality ablation, maintaining optimal performance when audio features are removed, compared to the heuristic approach’s equivalent retention. This differential robustness can be attributed to the cross-modal attention mechanism’s capacity for dynamic re-weighting of inter-modal relationships.

To complement the architectural analysis, Table 8 systematically addresses each research question mentioned in Section 1.2, demonstrating how our findings directly resolve the core challenges in behaviour-aware multimodal video summarization.

Table 8. Research Questions (RQs) Addressed in Behaviour-Aware Multimodal Video Summarization. This table systematically summarizes how each research question is resolved through our dual-framework approach, presenting core problems, solutions, key findings, and performance metrics.

RQs	Problem Assessed	Our Approach	Key Findings	Performance Results
RQ1: Behavioural cues extraction and cross-modal fusion	Existing summarization frameworks rely on unimodal or multimodal combinations, neglecting behavioural signals across multiple modalities [138, 84].	A multimodal feature extraction pipeline integrating MediaPipe [13], DeepFace [14], CLIP [15], RoBERTa [19], YAAPT [16], OpenSMILE [17], and HuBERT [18] with cross-modal attention [11].	Visual features provide the strongest standalone performance (F1-Score: 0.7986), while cross-modal attention emerges as architecturally critical as its removal precipitates a substantial decrease in F1-score.	Transformer-based approach: F1-Score of 0.8107, ROUGE-1 of 0.6765, representing 33.2% improvement over CLIP-It [7] and 7.3% enhancement over ScalingUp [6].
RQ2: Temporal alignment for coherence	Poor modality alignment disrupts narrative flow and summary coherence [6, 20].	Temporal alignment pipeline with Montreal Forced Aligner (MFA) [134], complemented by Whisper [142], and <i>bonus words</i> and cross-modal attention-based frameworks for synchronization.	The transformer-based approach demonstrates superior temporal consistency (Kendall’s τ : 0.6473) compared to the heuristic and other state-of-the-art approaches.	F1-Score: 0.8107, Kendall’s τ : 0.6473, Spearman’s ρ : 0.6466, CLIPScore: 0.5173 representing significant improvement over baseline methodologies.
RQ3: LLM adaptation for pseudo-ground truth generation	Dataset scarcity limits training while manual annotation remains expensive and subjective, necessitating scalable reference generation methodologies [23, 6].	We employ prompt-tuned LLMs for extractive summarization that preserves behavioural salience cues and temporal timestamps, enabling automated pseudo-ground truth generation for scalable evaluation without human annotation.	GPT-4.5 [161] demonstrates optimal reference quality with 93% thematic content retention and robust consistency across multiple generations (83% Jaccard similarity).	F1-Score: 0.8107 for GPT-4.5, 0.6823 for LLaMA-3.2.
RQ4: Evaluation metrics for behaviour-aware multimodal summarization	Traditional evaluation frameworks prove inadequate for assessing multimodal summary coherence due to segmentation biases and inability to capture cross-modal alignment quality [9].	A comprehensive evaluation suite incorporating ROUGE variants [24], BLEU [165], BERTScore [25], F1-Score, CLIPScore [26], Kendall’s τ , and Spearman’s ρ .	Transformer approach achieves superior visual-semantic alignment while heuristic excels in semantic fidelity, demonstrating complementary strengths across evaluation dimensions.	Transformer: BERTScore 0.9247, CLIPScore 0.5173. Heuristic: BERTScore 0.9536, CLIPScore 0.4162. Both outperform baseline methods across multiple metrics.

Limitations. Despite the substantial advances presented in this thesis, several limitations define the boundaries of our contribution and highlight areas requiring further investigation. The most significant constraint lies in our framework’s optimization for single-speaker interview videos in controlled settings, as exemplified by the ChaLearn First Impressions dataset. This specialization limits immediate applicability to multi-speaker scenarios, where overlapping speech, complex turn-taking dynamics, and multiple visual foci present challenges our current architecture cannot adequately address. Dynamic backgrounds, camera movements, and unstructured recording environments further complicate the behavioural feature extraction pipeline, potentially degrading performance in real-world applications beyond controlled interview settings.

The reliance on LLM-generated pseudo-ground truth summaries, while enabling scalable evaluation, introduces inherent biases into our assessment methodology. Large language models tend to prioritize semantically dense sentences over emotionally nuanced content with subtle behavioural cues, potentially misaligning with human preferences for summary quality. This limitation becomes particularly pronounced for short videos (under 10 seconds), where extractive summarization struggles to identify meaningful segments, requiring exclusion of such content from our evaluation framework. Additionally, the computational cost and API dependencies of LLM-based evaluation may limit reproducibility and accessibility for researchers with constrained resources.

Technical limitations manifest in several aspects of our implementation. Temporal misalignments between modalities affect approximately 9% of generated summaries despite our forced alignment approach, creating unnatural transitions that disrupt viewing experience. These misalignments stem from inherent challenges in audio-visual synchronization, transcription inaccuracies, and the discrete nature of our frame-based processing pipeline. The cross-modal attention mechanism, while generally effective, occasionally exhibits over-reliance on visual features when they present high confidence scores, leading to summaries that miss semantically significant speech content accompanied by subtle behavioural cues. Furthermore, our decoder’s dynamic length determination, while avoiding rigid compression ratios, sometimes produces summaries with suboptimal length—either too concise, omitting contextual information, or too detailed, including marginally relevant segments.

Future Work. Building upon the foundation established in this thesis, several promising research directions emerge that could substantially advance the field of behaviour-aware video summarization. The incorporation of additional behavioural modalities represents an immediate opportunity for enhancement. Eye gaze patterns, captured through modern eye-tracking technology or estimated through computer vision, could provide valuable insights into speaker attention and emphasis. Body posture analysis, extending beyond facial landmarks to full skeletal tracking, would capture broader gestural communication particularly relevant in presentation or lecture contexts. Micro-expressions, detected through high-frequency video analysis, could reveal subtle emotional states that current methods overlook.

Furthermore, the development of adaptive modality balancing techniques could optimize cross-modal integration based on content characteristics and modality reliability. Dynamic attention weighting mechanisms that adjust fusion parameters

according to modality confidence scores could prevent over-reliance on any single information source. Modality dropout during training, similar to standard dropout but applied at the modality level, could enhance robustness to missing or corrupted inputs. Learnable modality fusion layers, potentially implemented through neural architecture search, could discover optimal integration strategies beyond our current fixed architectures.

Expanding the application scope beyond single-speaker interviews represents a crucial direction for generalization. Multi-speaker scenarios require innovations in speaker diarization, visual attention modeling, and interaction dynamics understanding. Educational content, including lectures and tutorials, demands sensitivity to pedagogical structures and learning objectives. Unstructured content from social media platforms presents challenges in quality variation, background noise, and diverse recording conditions. Each domain requires specialized adaptations while maintaining the core principles of behaviour-aware summarization. Integration with large-scale pre-trained vision-language models and audio foundation models could provide more robust feature representations, particularly for out-of-domain content.

The exploration of advanced decoder architectures offers pathways to improved temporal modeling and summary coherence. Memory-augmented transformers could better capture long-range dependencies across extended videos, while graph-based decoders could explicitly model inter-segment relationships and narrative structures. Hierarchical decoding strategies might enable multi-granularity summarization, producing both brief highlights and detailed summaries from the same model. End-to-end audio-visual synchronization models could address current temporal alignment limitations, potentially eliminating the need for separate forced alignment steps.

Broader Impact. The development of behaviour-aware multimodal video summarization technology carries significant implications for society, with both transformative benefits and important considerations for responsible deployment. In educational contexts, our framework democratizes access to knowledge by enabling efficient consumption of lecture content, particularly benefiting students with learning differences, time constraints, or language barriers. The preservation of instructor emphasis and behavioural cues in summaries maintains pedagogical intent while reducing cognitive load, potentially improving learning outcomes and educational equity. Professional environments stand to gain substantial productivity improvements through automated meeting summarization that captures not just spoken content but also emotional dynamics and emphasis patterns, enabling better decision-making and reducing the burden of manual note-taking.

The accessibility implications of our technology are particularly noteworthy. For individuals with attention-related challenges, cognitive processing differences, or time constraints due to care-giving or work responsibilities, behaviour-aware summaries provide equitable access to video content that might otherwise remain inaccessible. The multimodal nature of our approach ensures that important information conveyed through any channel-visual, audio, or textual-is captured and preserved, benefiting users with sensory impairments who might rely more heavily on specific modalities.

From a sustainability perspective, our work contributes to more efficient digital content consumption, potentially reducing unnecessary computational resources spent

on streaming and reviewing full-length videos. By enabling viewers to quickly identify and access relevant content segments, we may contribute to reduced energy consumption in data centers and network infrastructure. This aligns with United Nations Sustainable Development Goals, particularly SDG 4 (Quality Education), SDG 8 (Decent Work and Economic Growth), SDG 9 (Industry, Innovation and Infrastructure), and SDG 12 (Responsible Consumption and Production).

However, we acknowledge potential risks and ethical considerations that must guide responsible development and deployment. Privacy concerns arise from the detailed behavioural analysis our system performs, necessitating robust data protection measures and transparent consent mechanisms. The risk of misuse through manipulated summaries that misrepresent speaker intent requires technical safeguards and ethical guidelines for implementation. Biases in training data or LLM-generated evaluations could perpetuate or amplify societal inequalities, demanding diverse dataset curation and bias mitigation strategies. The potential for surveillance applications of behavioural analysis technology raises important questions about individual autonomy and consent that the research community must address collectively.

6. CONCLUSION

This thesis focuses on developing behaviour-aware multimodal video summarization frameworks that capture both semantic content and behavioural nuances from human communication. A major challenge addressed in this study is the limitation of existing video summarization approaches that neglect behavioural cues conveyed through synchronized audio, visual, and textual modalities, failing to preserve the communicative intent and emotional context crucial for human-centric videos.

To address this challenge, we propose two complementary approaches: a transformer-based architecture with cross-modal attention mechanism and a heuristic method based on multimodal *bonus words* detection. The transformer-based framework integrates CLIP visual embeddings enhanced with facial movements and emotional transitions, HuBERT embeddings enriched with prosodic patterns, and RoBERTa textual embeddings. The cross-modal attention mechanism enables dynamic weighting of inter-modal relationships, while autoregressive decoding ensures temporal coherence in generated summaries. The heuristic approach demonstrates that meaningful improvements can be achieved through principled identification of terms that coincide with significant behavioural cues across modalities, providing an interpretable alternative with competitive performance.

The combination of behaviour-aware feature extraction and cross-modal fusion results in summaries that better preserve speaker intent, maintain narrative coherence, and capture emotionally significant moments that traditional methods overlook. Our comprehensive ablation studies reveal that visual features provide the strongest stand-alone performance, yet optimal results require synergistic integration of all modalities. Most significantly, the removal of cross-modal attention mechanisms causes performance degradation than removing any individual modality, establishing that inter-modal relationship modeling is as crucial as the modalities themselves.

In conclusion, this thesis establishes behaviour-aware multimodal video summarization as a significant advancement in multimedia understanding, with demonstrated improvements across multiple evaluation dimensions and clear pathways for future development. By integrating visual, audio, and textual modalities through sophisticated architectural mechanisms, we have shown that machine-generated summaries can preserve the rich behavioural nuances of human communication. While limitations exist and challenges remain, the principles and methods developed here provide a foundation for continued progress toward more nuanced, equitable, and human-centric approaches to video content understanding in our increasingly multimedia-mediated world.

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